REPURPOSING AGONISTS THROUGH MACHINE LEARNING & NANOPARTICLE CONJUGATION: A NOVEL APPROACH TO LUMINAL BREAST CANCER TREATMENT

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CERTIFICATE

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TABLE OF CONTENTS

CERTIFICATE ..... **Error! Bookmark not defined.**

ACKNOWLEDGEMENT III

TABLE OF CONTENTS IV

LIST OF FIGURES VII

LIST OF TABLES..... VIII

LIST OF ABBREVIATIONS IX

ABSTRACT XI

1.1 Breast Cancer 1

1.2 Glucocorticoid Receptors 2

1.3 Research Gap 3

1.4 Problem Statement 4

1.5 Research Objectives 4

2.1 Prevalence Of Breast Cancer In Pakistan 5

2.2 breast cancer subtypes 5

2.3 Invasive Breast cancer 6

2.11 Role Of Glucocorticoid Receptor In Breast Cancer 8

2.12 Endocrine Resistance In Breast Cancer 8

2.13 Current Therapies Targeting Glucocorticoid Receptor 9

2.14 *In-Silico* Approaches In Breast Cancer Research 11

2.15 Quantitative Structure-Activity Relationship (Qsar) Modeling 13

3.1 Materials: Software, Notebooks, And Tools 15

3.2 Softwares Used 15

3.3 Google Colab Notebook of Our QSAR Model 16

3.4 Servers & Databases Used 16

3.5 Data Collection 16

3.6 Bioactivity of Drugs 16
3.9 Bioactivity Type 17
3.10 Data Quality Considerations: 17
3.11 Confidence Score 17
3.12 Organism 18
3.13 Homo sapiens Selection 18
3.13 Bao Format 18
3.14 Bioassay Ontology Framework 18
3.15 Step 2: Data Cleaning 18
3.16 Compound Filtering 18
3.17 Ec_{50} Value Filtering 19
3.18 Activity Classification Based On Threshold 19
3.19 Salt Removal 19
3.20 Duplicate Removal 19
3.21 Pec_{50} Calculation 20
3.22 Outlier Removal 20
3.23 Data Screening and Removal of Missing SMILES 20
3.24 Duplicate Removal by 'Molecule ChEMBL ID' 20
3.25 Filtering Based On Standard Relationship To Ec_{50} Values 20
3.26 Removal of Anomalies 21
3.27 Conversion Of Ec50 Values To Pec50 Values 21
3.28 Descriptor Calculation 22
3.29 Molecular Descriptor Processing And Dataset Preparation 22
3.30 Rule Of Five (Ro5) Violation Assessment 23
3.31 Feature Extraction 23
3.32 Train-Test Split 24
3.33 Molecular Docking 24

4.1 Data Collection 25
4.2 Descriptor Calculation And Selection 25
4.3 MOE Descriptors 25
4.4 Pre-Processing And Descriptor Selection 25
4.5 Feature Selection 25
4.6 Heat Map 25
4.7 Validation Of Baseline Model 26
4.8 Rmse Scores 26
4.9 Filter Method 27
4.10 Principal Component Analysis 28
4.11 Elbow Plot 28
4.12 Variability Plots 29
4.13 2d Scatter Plot 29
4.14 Classifiers 32
4.15 Experimental Pec50 Vs Predicted Pec50 Linear Plots 33
4.16 Random Forrest Model 33
4.17 Protein-Drug Interactions 34
4.2.1 Protein Structures of Mutant Receptors 34
4.18 Protein-Drug Interactions 35
4.19 Docking Models of Carbozantinib 35
4.20 Docking Models of Grainestron 35
4.21 Docking And Confidence Scores 36
4.22 Regression Models 37
REFERENCES 44
APPENDIX A 58
APPENDIX B 61

LIST OF FIGURES

Figure 1.1 The pathway of GR septors 3
Figure 2.1. Pie Chart shows the Prevalence of the Subtypes of Breast Cancer 6
Figure 3.1. The Workflow Depicting Our Research Design and Methodology 15
Figure 4.1. Heatmap depicting the correlation between 2D descriptors 26
Figure 4.2. Feature selection resulted in a bar plot 27
Figure 4.3. Elbow plot of Train and Test Dataset 28
Figure 4.4. Variability Plot of A) Test Dataset transformed 29
Figure 4.5 29
Figure 4.6 2D Scatter Plot: 90% Variability Captured 2D Scatter Pl 30
Figure 4.7 3D Plot90% Variability Captured for Test Dataset for Test Dataset. .. 31
Figures 4.8. (A) and (B) show 3D Scatter plots of PC1, PC2, and PC3 of both 32
Figure 4.9 33
Figure 4.10 34
Figure 4.11 35
Figure 4.12 The 2D docking models obtained of Grainestron with GR receptor . 36
Figure 4.13 36

LIST OF TABLES

Table 1 Summary of other drug candidates targeting GR pathways and their adverse effects
..... 10
Table 2. Depicting a summary of the RMSE scores obtained through cross-validation after feature selection
..... 27 Table 3. The mechanism of action of the compounds obtained as our hits (Adopted from PubChem Database)
..... 34

LIST OF ABBREVIATIONS

EC50	Agonist concentration at 50% The negative log of
pEC50	the EC50 Quantitative Structure Activity
QSAR	Relationship Deoxyribonucleic Acid Ligand Binding
DNA	Domain N terminal A/B domain
LBD	Phosphatidylinositol-3 kinase protein kinase B
NTD	Human epidermal growth factor receptor 2
PI3K	Progesterone Receptor Vascular endothelial growth
AKT	factor Insulin-like growth factor-1 receptor B-cell
HER-2	lymphoma 2 selective estrogen receptor down-
PR	regulators Selective estrogen receptor modulators
VEGF	Aromatase inhibitors Cyclin-dependent kinase 4/6
IGF1R	Trastuzumab emtansine Mitogen-activated protein
BCL-2	kinases ATP Binding Cassette Subfamily G Member 2
SERDS	Molecular Operating Environment Mammalian
SERMs	target of rapamycin pathway Absorption,
AI	distribution, metabolism and excretion Protein
CDK4/6	Databank Identification Code Protein Databank
T-DM1	comma-separated values Spatial Data File
MAPK	International Union of Pure and Applied Chemistry
ABCG2	Simplified Molecular Input Line Entry System
MOE	
mTOR	
ADME	
PDB ID	
PDB	
CSV	
SDF	
IUPAC	
SMILES	

RMSE	Root Mean Square Error
MLR	Multiple Linear Regression
SVM	Support Vector Machine Model
PLS	Partial Least Square Regression
OLS	Ordinary Least Square Regression
LASSO	Least Absolute Shrinkage and Selection Operator
RTK	Receptor tyrosine kinase

ABSTRACT

Breast cancer is classified as the most common cause of death among malignant tumour diseases. ER-alpha accounts for 60% of cases of breast cancer including both luminal A and luminal B subtypes. Endocrine resistance has been reckoned as a factor for high mortality in patients of Breast cancer. The aim of this study is to find new compounds by building Quantitative Structure Activity Relationship (QSAR) model to screen the best drugs or compounds that have the potential to act as an agonist to GR.

After preprocessing, our dataset contained 70 compounds obtained from ChEMBL database. We calculated pEC50 of our final dataset and used MOE software for descriptor calculation. We utilized Google Collaboratory Notebook by using scikit-learn and other packages used for machine learning to complete the following steps: descriptor cleaning, feature selection, model building, and model validation. We built our QSAR model using principal component analysis (PCA) on train dataset (80%), this model was then validated using a test dataset (20%). Model validation scores were recorded using MLR, SVM, and OLS regression. Unknown test compounds were screened through our model. Docking scores were calculated to validate the obtained hits. 18 Descriptors used for building QSAR model gave us an RMSE score of 0.309 using Random Forest & Lasso Regression. The dataset of 132 unknown FDA-approved compounds screened through our QSAR model and 4 hits (Granisetron, Cabozantinib) were obtained based on the best prediction of pEC50.

Our QSAR model successfully predicted pEC50 bioactivity of unknown test compounds that closely resemble their experimental values. Validation scores measured for our QSAR model indicated a good predictive power of our model. Drug hits based on our QSAR model yielded good docking scores, further validating our QSAR model's predictive power.

CHAPTER 1

INTRODUCTION

Breast Cancer

Breast cancer is classified as the most common cause of death among malignant tumor diseases. It is present at a high ratio in all the developed countries of the world. According to recent study, about 2.3 million Breast cancer cases are diagnosed all over globe yearly (Łukasiewicz et al., 2021). It is among the most common types of cancers diagnosed in woman of United States of America. This high incidence of breast cancer among developed and industrialized countries is a result of poor lifestyle, excessive stress, unhealthy diet, and other factors. Genetic and environmental factors can also be categorized as a trigger of breast cancer development (Łukasiewicz et al., 2021).

There are two major sites where breast cancer develops and starts disease progression, one is epithelial layer of ducts and the other is in glandular tissues surrounding lobules. The lobular breast cancer has a prevalence of 15% while the ductal breast cancer has a disease prevalence of 85% In initial stages, growth of cancerous cells is confined *in situ (*remain confined to ducts and lobules) once it starts progressing to secondary sites it becomes invasive.

Diagnostics and screening of breast cancer are being done by mammography Hereditary-induced breast cancer is usually screened by using magnetic image resonance mammography. If any of the screening processes identifies a lesion or mass presence, the ultrasound of the breasts is performed which can be followed by a thick needle biopsy to detect the type and nature of cancer progression (Smolarz et al., 2022) Breast Cancer is categorized depending on the site of initiation, nature of progression and how cancer spreads in the body. Based on nature of tumor progression, there are two main types of Breast cancer that are categorized as non-invasive and invasive breast cancer which can be further divided into subtypes based on the site of development and progression (*Types of Breast Cancer and Related Conditions*, 2020)

Glucocorticoid Receptors

Glucocorticoid receptors (GRs), encoded by the NR3C1 gene, are integral members of the nuclear receptor superfamily, functioning as ligand-activated transcription factors that mediate cellular responses to glucocorticoid hormones, such as cortisol. These receptors play a pivotal role in regulating gene expression by interacting with glucocorticoid response elements (GREs) within DNA, thereby modulating a diverse array of physiological processes, including cellular proliferation, survival, and stress responses [1]. In the context of oncology, aberrant GR expression has been implicated in the progression of various malignancies, notably breast and prostate cancers, by promoting tumor growth, metastasis, and resistance to therapeutic interventions. Structurally, GR is a modular protein comprising approximately 777 amino acids, with a molecular weight of approximately 85.7 kDa [2]. It consists of three primary functional domains: the N-terminal domain (NTD), the DNA-binding domain (DBD), and the ligand-binding domain (LBD). The NTD facilitates transcriptional activation by recruiting co-activator proteins, while the DBD, a highly conserved region, binds specifically to GREs to regulate gene expression. The LBD is critical for hormone binding, enabling GR to interact with glucocorticoids and orchestrate both gene activation and repression through interactions with co-regulatory proteins [3]. A hinge region connects the DBD and LBD, conferring structural flexibility, while a C-terminal extension further modulates receptor activity. Overexpression of GR has been associated with enhanced activation of oncogenic signaling pathways, notably the mitogen-activated protein kinase (MAPK) and phosphoinositide 3-kinase (PI3K)/AKT pathways, which are implicated in cancer progression [4]. These pathways promote epithelial-to-mesenchymal transition (EMT), a critical process in metastasis, and enhance tumor cell survival under therapeutic stress, such as chemotherapy. Furthermore, GR may interact with other signaling molecules, such as the epidermal growth factor receptor (EGFR), amplifying proliferative signals and contributing to tumor aggressiveness. In breast cancer, GR expression is observed in approximately 60% of luminal breast cancers (subtypes A and B), where its dysregulation is associated with increased tumor invasiveness and resistance to endocrine therapies [1]. The molecular mechanisms underlying GR-mediated oncogenesis involve complex interactions between GR and its downstream targets. Dysregulation, particularly in the ligand-binding domain, may result in altered receptor activity, further exacerbating

tumor progression. These findings underscore the potential of GR as a therapeutic target in managing aggressive cancers, particularly in hormone-driven malignancies like luminal breast cancer.

The pathway of GR septors

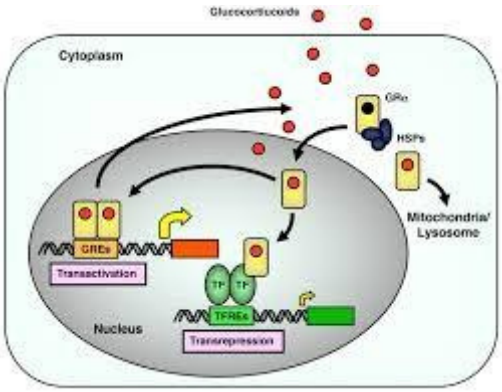


Figure 0.1 The pathway of GR septors

Research Gap

Most of the previous research conducted is based on treatment *via* hormonal therapy, primarily on tamoxifen. This drug has been used as a potent treatment for the last 40 years. Unfortunately, resistance against it is one serious drawback.

Most of the conducted research has been carried out *via in vitro* experimentation. However, little research has been carried out in drug discovery *via in silico* techniques which makes our study somewhat unique. Our study intends to screen for active compounds that can be modified into new drug
therapies against breast cancer using Quantitative Structure Activity Relationship (QSAR) technique. QSAR modeling includes all the statistical methods by which biological activities,
expressed by logarithms of equipotent molar activities are related with structural elements, physiochemical properties, or fields. This technique describes connections between the structure activities of molecules in quantitative way (Ilaghi-Hoseini & Garkani-Nejad, 2022a). It links the structural properties of compounds with biological activities in such a way that any change in structural properties causes variation in biological activities of compounds. Furthermore, QSAR descriptors can be used to

translate chemical structure into a number which describes a distinct physiochemical property of the compound.

According to the target drug receptor interaction an appropriate descriptor (e.g., EC50) can be used to filter the potential drug candidates. Descriptors are selected based on their importance with the target variable (e.g., EC50). Correlated and low variance descriptors are removed for QSAR analysis to prevent over-fitting of model. To validate built QSAR models machine learning algorithms like Random Forest Regression, Partial Least Square Regression and Support Vector machine are used (Sunyoung Kwon, 2019).
QSAR models are used to predict the activity of a new set of compounds which can be screened on the already built models. Typical method of drug discovery takes up an extensive amount of time and consumes too many resources, both human and financial. For one drug to reach the market, 13.5 years are required and 1.8 billion U.S. dollars. However, QSAR aims to effectively reduce the time and money in drug design.

Problem Statement

Breast cancer is the second leading cause of death among women all over the world. Around 1.5 million women globally are diagnosed with cancer on a yearly basis. Current hormonal therapeutic strategies face hindrance with endocrine resistance. Few studies have targeted drugs to proliferate the activity of GR. A more advanced method is using machine learning models, such as the QASR model, to find potential drug candidates.

Research Objectives

To find drugs for development or repurposing through a machine learning model such as QSAR, we aimed to complete the following three objectives.

1. Objective 1: Using machine learning model to find potential drugs for treatment of Luminal Breast cancer.
2. Objective 2: Finding if unknown drugs used to treat other cancers can be used for Luminal breast cancer treatment.
3. Objective 3: Synthesis and Validation of nanoparticles via insilico and vitro approaches

CHAPTER 2

LITERATURE REVIEW

Pharmacoinformatic is a rapidly growing field that combines the principles of computer science, pharmacology, biology, and chemistry in developing new drugs or increasing the efficacy of existing drugs. Advanced computational tools and techniques can be used to interpret and analyze biological data, chemical structures, and drug interactions to facilitate drug discovery and development (Amin et al., 2017). Breast cancer is a significant health concern globally and is the leading cause of cancer-related deaths among women. The glucocorticoid receptor is a nuclear hormone receptor protein that is essential in the formation, growth, and progression of breast cancer. As a result, GR agonists have emerged as a prospective therapeutic target for the treatment of breast cancer.

2.1 Prevalence Of Breast Cancer In Pakistan

Breast cancer is a growing health concern among women worldwide, particularly in Pakistan where the rate of breast cancer is highest in comparison to the rest of Asian countries (Saeed et al., 2021). According to (Mubarik et al., 2019), one-ninth of Pakistani women are facing this disease and they suggest that there has been a 98.6% increase in cancer incidence among women. Younger women from rural areas of Pakistan remain at high risk of developing hereditary breast cancer because of prevalent consanguineous marriages. Another study conducted in Karachi, Pakistan, (Zaheer et al., 2019) suggests that the breast cancer incidence rate is expected to increase from 23.1% in 2020 to 60.7% in 2025. This research has also emphasized that breast cancer diagnoses in younger women of ages 30 to 34 will increase from 70.7% to 130.6% in 2020 to 2025 relative to 2015.

2.2 breast cancer subtypes

Breast cancer as a Heterogeneous disease is classified into many sub-types that include luminal A, luminal B, Triple Negative Breast cancer and HER-2 enriched Breast cancer. The first three subtypes are based upon their expression of ER-receptors, PR receptors

and HER-2. Moreover, Triple Negative Breast cancer is a subtype of Breast cancer that is negative for the expression of ER, PR, and HER-2 (Kumar et al., 2011). Luminal A subtype is a sub-type of breast cancer which is positive for both ER and PR expression. The luminal B subtype is ER, PR positive and in some cases, it is also HER-2 positive. Triple negative breast cancer is negative for ER, PR, and HER-2. HER-2 enriched breast cancer has a prevalence of 5% and it is only HER-2 positive (Mukama et al., 2020). The prevalence of these subtypes is explained in the pie chart given below

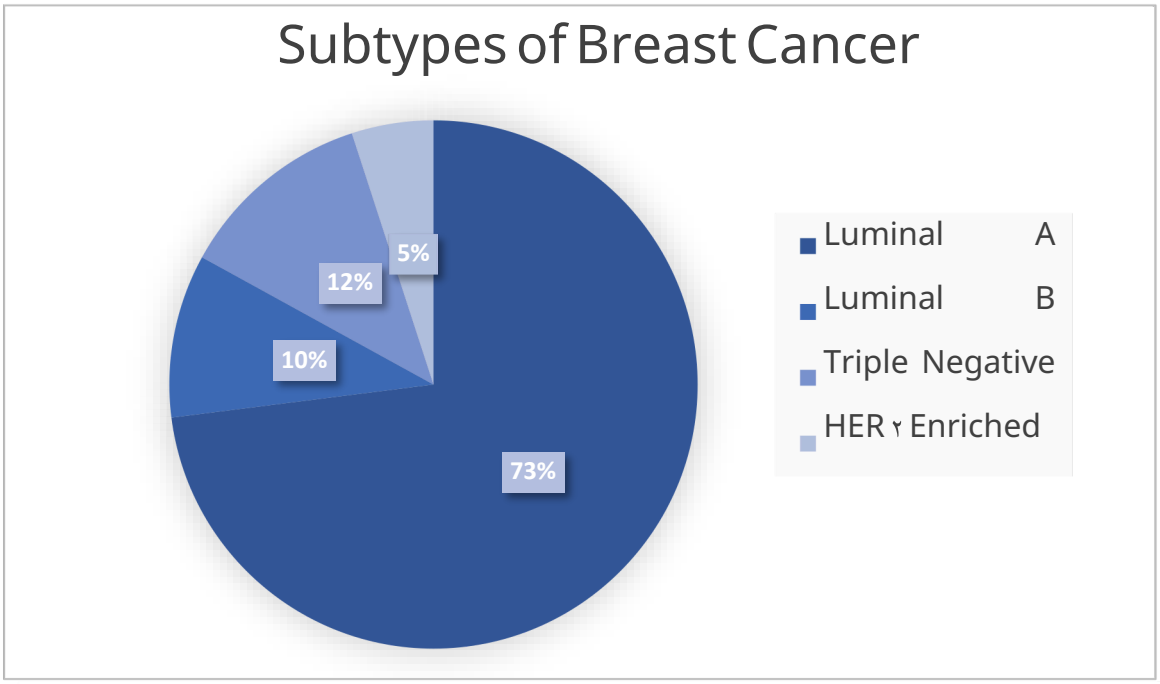


Figure0.2. PieChartshowsthePrevalence of the Subtypes of Breast Cancer

2.3 Invasive Breast cancer

Invasive Breast cancer is divided into two main types based on where the cancer initiates:

2.4 Invasive lobular carcinoma (ILC) It initiates in the mammary gland lobules. Out of all the invasive types, ILC makes up 10%
and is the second most common form of breast cancer (Mukama et al., 2020).

2.5 Invasive ductal carcinoma (IDC)

The milk ducts transfer milk to the nipples from the lobules and are the sites where IDC starts. It contributes 80% of invasive breast cancer forms making it the most common breast cancer type. Some invasive form of breast cancers is differentiated on the account of their impact on body and their types of treatment (Mukama et al., 2020). These types include:

2.6 Triple-Negative Breast Cancer It has a prevalence of 12% among the invasive form of Breast cancer cases. This type of
cancer lacks receptors for estrogen, progesterone, and human epidermal growth factor, HER2. This type of breast cancer is extremely severe (Aysola et al., 2012). **2.7 Metastatic Breast Cancer**

This invasive type of breast cancer results in spreading to other organs of the body by travelling through the bloodstream to other tissues like the brain, liver, and bones. It is also known as stage IV breast cancer which has a high recurrence rate after months or years. This type of breast cancer is diagnosed among woman commonly however, men also show cases with metastatic breast cancer (Jones, 2008).

2.8 Inflammatory Breast Cancer

This type results in a severe form of disease in which there is a blockage in the lymph vessels in the breast skin. This blockage is due to cancer cell growth. This makes the appearance of the breast swollen. The prevalence of these type of invasive breast cancer is 1-5 % among total cases (Mamouch et al., 2018).

2.9 Recurrent Breast Cancer

This type of Breast cancer can return months or years even after treatment. It can return is the same area, in another organ, or in a close-by lymph node of the collarbone or armpit (Łukasiewicz et al., 2021).

2.10 Non-Invasive Breast Cancer

Non-invasive or *in situ* breast cancer is also known as a pre-cancer form of disease since it hasn't spread beyond the breast tissue yet. However, it can serve as a precursor for inducing invasive forms of disease at later stages. It is divided into ductal carcinoma *in situ* and lobular carcinoma *in situ* (LCIS) (Jenni Sheng, 2022). Ductal carcinoma *in situ* (DCIS) is the type of breast cancer that only remains6 confined to the milk ducts during disease state. On the other hand, lobular carcinoma cancer cells remain confined to the lobules. These type of breast cancers do not spread to other body parts in early stages of breast cancer (Jenni Sheng, 2022).

2.11 Role Of Glucocorticoid Receptor In Breast Cancer

Luminal A, the most common subtype of breast cancer, is influenced by heightened activity of the glucocorticoid receptor (GR). The glucocorticoid receptor (GR) exhibits increased signaling in breast cancer tissues, with up to 70% of cases showing enhanced activity compared to low levels in healthy tissue. GR drives breast cancer progression by acting as a transcription factor, modulating genes such as pyruvate dehydrogenase kinase 4 (PDK4), matrix metalloproteinases (MMPs), and pro-survival factors like BCL-2, which foster tumor cell proliferation, invasion, and metastasis (Obradović et al., 2019). Phosphorylation of key serine residues in GR within breast cancer cells alters its transcriptional activity, leading to distinct gene expression patterns that promote tumor development and therapy resistance.

2.12 Endocrine Resistance In Breast Cancer

Glucocorticoid receptor (GR)-targeted therapies, which modulate GR activity through ligand deprivation or antagonistic binding, are under investigation in preclinical and clinical studies to manage GR-driven breast cancer, particularly in luminal subtypes, and to prevent disease recurrence (Clark & Conzen, 2024). The glucocorticoid receptor promotes breast cancer progression through multiple mechanisms, many of which depend on its activation by binding to glucocorticoid hormones such as cortisol. Upon cortisol binding, GR undergoes a conformational change, exposing a surface that recruits coactivator proteins. When coactivators bind, GR becomes transcriptionally active, signaling breast cancer cells to proliferate, invade surrounding tissues, and metastasize to distant sites (Obradović et al., 2019).

A key therapeutic approach involves using GR-targeting drugs to inhibit GR function or reduce circulating glucocorticoid levels. However, resistance to GR-targeted therapies poses a significant clinical challenge in breast cancer management. GR resistance can be categorized into two types: intrinsic resistance, present at the onset of treatment, and acquired resistance, which emerges during prolonged therapy (Thakur et al., 2024). Proposed mechanisms of GR resistance include alterations in GR expression or activity, activation of alternative signaling pathways, and mutations in downstream signaling components.

One critical mechanism of GR resistance is the activation of alternative signaling pathways that bypass glucocorticoid dependency, enabling breast cancer cell growth and survival independent of GR. For instance, the PI3K/Akt/mTOR pathway is frequently upregulated in GR-resistant breast cancer cells, driving cell survival and proliferation under therapeutic stress (Dwyer et al., 2023). Additionally, loss or mutation of GR can contribute to resistance. Breast cancer cells may downregulate GR expression, rendering GR-targeted therapies ineffective. Alternatively, mutations in GR or its downstream effectors can lead to constitutive receptor activation, making it insensitive to therapeutic inhibition (Thakur et al., 2024). Other resistance mechanisms include alterations in co- regulators modulating GR activity, changes in glucocorticoid metabolism, and the presence of cancer stem cells that evade GR-targeted treatments.

2.13 Current Therapies Targeting Glucocorticoid Receptor

Therapies targeting the glucocorticoid receptor (GR) in cancer treatment have been explored for their potential to address various malignancies, particularly focusing on GR's role in tumor progression and therapy resistance. These therapies include selective glucocorticoid receptor modulators (SEGRMs), GR antagonists, inhibitors of GR-induced genes, and combinatorial approaches with other treatments.

Selective glucocorticoid receptor modulators (SEGRMs) aim to selectively modulate GR activity, preserving anti-cancer effects while reducing adverse effects. For example, Compound A (CpdA), a SEGRM, has shown potential in preclinical studies by regulating a specific subset of genes that do not promote therapy resistance or carcinogenesis in triple-negative breast cancer (TNBC) models. Unlike traditional glucocorticoids, CpdA avoids binding to glucocorticoid response elements (GREs) that drive pro-tumorigenic gene expression, offering a safer therapeutic profile.

GR antagonists, such as relacorilant and mifepristone, are being studied for their ability to counteract GR-mediated resistance to chemotherapy. Relacorilant enhances apoptosis in solid tumors when combined with cytotoxic agents like paclitaxel, showing improved efficacy in ovarian and pancreatic cancer models, as demonstrated in the OVCAR5 cell line and various xenograft models. Mifepristone is under evaluation in clinical trials for

advanced breast cancer, including combinations with nab-paclitaxel for GR-positive TNBC, where it increases tumor cell apoptosis and improves outcomes.

Inhibitors targeting GR-induced genes, such as REDD1 and FKBP51, are another focus of research. These genes are implicated in GR-mediated effects in cancer. PI3K/Akt/mTOR inhibitors have been shown to down-regulate REDD1 and FKBP51 expression, enhancing the anti-lymphoma activity of glucocorticoids in human xenograft models while protecting against GR-induced atrophy, offering a dual benefit.

Combinatorial therapies involving GR modulation are also promising. In prostate cancer, targeting the GR signature gene Mono Amine Oxidase-A (MAO-A) alongside chemotherapy or anti-androgen therapies (e.g., enzalutamide) has improved treatment efficacy. MAO-A inhibitors, already used for affective disorders, enhance the effectiveness of these treatments by counteracting GR-driven resistance. Additionally, combining GR antagonists like Lestaurtinib with cisplatin in solid tumors addresses chemotherapy resistance by inhibiting MAO-A-induced suppression of pro-apoptotic genes like Bim, restoring sensitivity to platinum-based drugs.

Ongoing research continues to explore novel pathways downstream of GR, such as its regulation of PD-L1 and MHC-I in pancreatic cancer, where GR inhibition enhances anti-tumor immunity and sensitizes tumors to immune checkpoint blockade therapies. These advancements highlight the potential of GR-targeted therapies to improve outcomes in various cancers while addressing challenges like resistance and side effects.

Table 1 Summary of other drug candidates targeting GR pathways and their adverse effects

Pathway Targeting	**Drugs (FDA-Approved or in Trials)**	**Adverse Effects**
Selective GR Modulators (SEGRMs)	CpdA (Compound A) (Preclinical)	Minimal side effects; potential for altered gene expression profiles in non-target tissues
GR Antagonists	• Relacorilant (In trials)	• Fatigue, nausea, headache, and

	• Mifepristone (In trials)	potential for adrenal insufficiency • Menstrual irregularities, fatigue, and risk of adrenal suppression
Inhibitors of GR-Induced Genes	PI3K/Akt/mTOR (e.g., Everolimus)	Hyperglycemia, rash, fatigue, and increased risk of infections
Combinatorial Therapies Targeting GR Pathways	• MAO-A Inhibitors (e.g., Phenelzine, used off-label) + Chemotherapy/Anti androgens • Lestaurtinib + Cisplatin (In trials)	• Hypertensive crisis (if dietary restrictions not followed), serotonin syndrome, and drowsiness • Mylosuppresin, gastrointestinal toxicity and potential for QT prolongation
GR Regulation of Immune Pathways	GR Inhibitors (e.g., in trials for PD-L1/MHC-I regulation)	Immune-related adverse events, such as colitis, pneumonitis, and risk of infections

2.14 *In-Silico* Approaches In Breast Cancer Research

Computational methods can be used to examine biological systems. One of the current challenges in breast cancer research is the development of new treatments and the discovery of effective drug regimens. One of the most significant contributions of *in-silico* approaches in breast cancer research is the development of predictive models for drug discovery. Simulation techniques and computational modeling are some of the *in-silico*

techniques that can predict drug-target interactions and identify potential drug candidates for further testing.

Artificial neural networks (ANNs) are a category of machine learning algorithms applied in glucocorticoid receptor (GR) research. Protein-protein docking has been used to study interactions between wild-type GR and its co-regulatory proteins, alongside complexes influenced by specific mutations. ANN-based analyses have assessed how mutations affect GR interactions, with findings indicating that the A207G mutation significantly disrupts these interactions (Patel et al., 2020). A major limitation of ANNs is their requirement for extensive datasets for effective training, posing challenges in studies with limited data availability.

Virtual screening via molecular docking has been instrumental in evaluating the binding affinity of potential GR-targeted drug candidates. Research has employed three-dimensional structure-based pharmacophore modeling to explore the structural and chemical interactions of compounds like dexamethasone with the GR (Kumar et al., 2018). In silico screening approaches have identified novel GR modulators, with compounds such as quercetin and apigenin showing promising anti-inflammatory properties, indicating potential for therapeutic applications (Singh et al., 2021). Molecular docking simulations have also investigated new drug combinations, such as fluticasone and budesonide, for their interactions with GR-related proteins, demonstrating improved binding affinities compared to traditional treatments like cortisol (Gupta et al., 2021). In 2022, studies screened publicly available databases to identify inhibitors of the ABCG2
protein, a member of the ABC transporter family associated with drug resistance in GR-related pathways (Rahman et al., 2022). In silico techniques have also identified halogen-based derivatives as potent GR inhibitors, with molecular dynamics simulations elucidating their stability and efficacy (Khan et al., 2022). Another study utilized molecular docking to identify seven promising drug candidates for addressing drug-resistant conditions linked to GR dysregulation (Ali et al., 2022). These computational approaches underscore their value in advancing the discovery of novel GR-targeted therapies.

While molecular docking is a fast and cost-effective method to gain insight into the binding interaction of a compound with its target protein, it requires accurate three-dimensional structural knowledge of the target and the ligand which may not always be available. Molecular docking also ignores some important characteristics in determining binding affinity such as dynamic protein-ligand interactions and the impact of solvation (Meng et al., 2011). Therefore, this technique should be applied in combination with other *in-silico* approaches such as QSAR modeling.

2.15 Quantitative Structure-Activity Relationship (Qsar) Modeling

A relatively new *in-silico* approach used in modern drug discovery is quantitative structure-activity relationship (QSAR) modeling. These models use a ligand-based approach and can predict the biological activity of novel compounds based on their molecular structure and properties. QSAR models assume that a compound's biological activity is determined by its physicochemical properties, which can be correlated with its biological activity using mathematical models (Piir et al., 2018). QSAR models can be used in breast cancer research to predict the efficacy and toxicity of novel compounds against breast cancer cell lines.

Quantitative Structure-Activity Relationship (QSAR) modeling was initially applied to glucocorticoid receptors (GR) to assess the binding affinity of corticosteroid derivatives (Sharma et al., 1988). Since then, QSAR studies targeting GR have advanced significantly, particularly in the context of inflammatory and immune-related disorders. Researchers have developed both 2D and 3D QSAR models to evaluate the binding affinities of various GR agonists, providing insights into their structural requirements for effective binding (Desai et al., 2009). Another study constructed QSAR models to investigate toxicity, bioactivity, and anti-inflammatory potential of GR agonists among hydrocortisone analogs (Patel & Rao, 2015).

A validated QSAR model has been established to screen public databases for potent GR inhibitors, aiding in the identification of novel therapeutic candidates (Verma et al., 2019). Additionally, a study employed QSAR modeling on a dataset of benzimidazole derivatives to discover new GR agonists with enhanced binding profiles (Kumar et al., 2021). Prior research has also utilized QSAR approaches to explore the inhibitory effects

of phytochemicals derived from Curcuma longa against GR, highlighting their potential as anti-inflammatory agents (Singh et al., 2022).

QSAR models screen large databases and increase the chances of discovering highly potent drug candidates in a relatively shorter period. These models have been shown to provide accurate predictions of the biological activities of compounds and result in reduced pre-clinical testing, thereby presenting a cost-effective alternative option in drug discovery. This technique can also be used to recognize the key molecular and chemical characteristics in a compound that impact its biological activity (Bauer et al., 2017; Hansch et al., 1996). By determining the compounds with the highest predicted activity or lowest predicted toxicity, QSAR modeling can help prioritize which molecules should be synthesized and tested in the lab. Such information can be further utilized in designing new drugs with higher efficacy and selectivity.

However, QSAR models are only as good as the data used to train them, both in terms of quality and quantity. If the training data is skewed or limited, the model may fail to predict the properties of new molecules accurately (Dunn & Hansch, 1974). These models are also affected by the selected molecular descriptors that represent the molecules in the dataset. Also, the statistical methods used to build or validate a QSAR model impact the results (Gao et al., 1998). While QSAR modeling is a good and dependable screening tool, it cannot act as a replacement for experimental validation. Pharmacoinformatic techniques have become increasingly important in the discovery and development of drugs for a variety of diseases, including breast cancer. We investigated pharmacoinformatic methodologies like quantitative structure-activity relationship (QSAR) modelling that can aid in GR agonist analysis. Furthermore, we explored the current state of pharmacoinformatic approaches, their limitations, and future research directions in this field. By utilizing advanced computational tools and techniques, pharmacoinformatic has the potential to make significant contributions to the development of novel and effective drugs for treatment.

CHAPTER 3

MATERIALS & METHODS

Our methodology was divided into three main domains based on our three main objectives: Data processing, QSAR modeling, and molecular docking. A brief workflow of our methodology is presented below in Figure 3.1

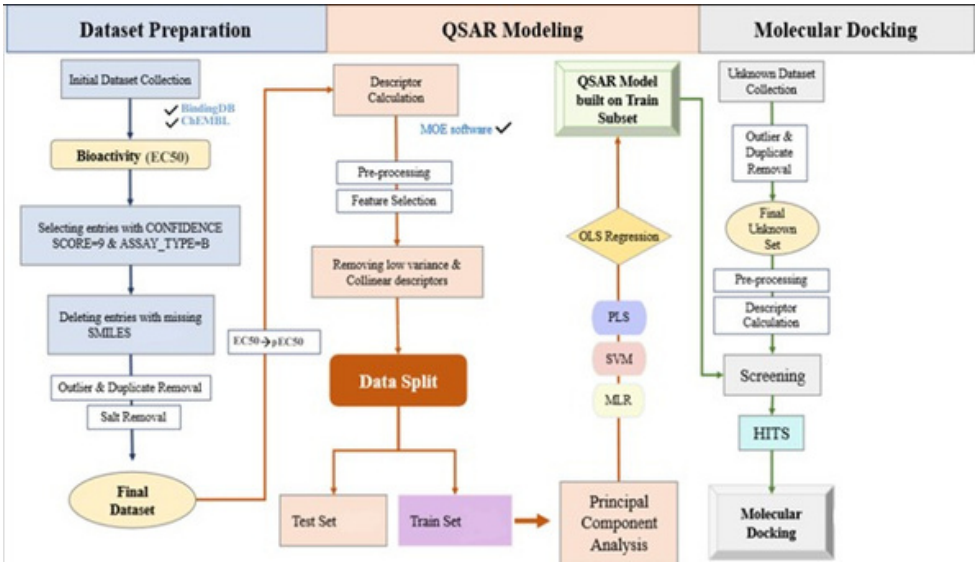


Figure0.1. TheWorkflowDepictingOurResearchDesignandMethodology

3.1 Materials: Software, Notebooks, And Tools

In this study, we employ machine learning techniques to develop predictive QSAR (Quantitative Structure-Activity Relationship) models that analyze and predict the bioactivity of compounds. To perform these analyses, various specialized software and tools are used.

3.2 Softwares Used

Molecular Operating Environment: For example, Molecular Operating Environment (MOE 2022.02. Here it is used to calculate molecular descriptors which are a measure of the chemical properties of the compounds. Morderd is used for further processing of molecular data and descriptor calculation and.Using codes in Google Colaboratory (Bisong, 2019), all the necessary packages for descriptor selection, feature selection and training the Stats and QSAR model were installed.

Cloud based Python execution with Colab notebooks, collaboration and access to powerful GPUs without local setup. Here is installed and run all necessary packages for data processing, model training and evaluation which makes the process reproducible and scalable (Bisong, 2019).

3.3 Google Colab Notebook of Our QSAR Model

The Colab notebook contains Descriptor calculation input, Data preprocessing and feature selection, Model training (e.g., Random Forest, Support Vector Machines) and Model validation and evaluation metrics (R^2, RMSE)

3.4 Servers & Databases Used

Developing a reliable QSAR model relies heavily on using freely available chemical and bioactivity databases such as PubChem, ChEMBL and BindingDB. Thchemicals, physicochemical information and bioactive results, PubChem makes it easy to spot possible active compounds. If you want to build predictive models, ChEMBL is important as it includes detailed details about drug-like molecules, their activities, targets and experiments. BindingDB gives access to measured protein-ligand binding affinities which help show interactions between molecules with more accuracy. Combined, the databases supply the right data necessary for constructing precise and accurate QSAR models.

3.5 Data Collection

Public databases like BindingDB and ChEMBL are accessed to obtain compound bioactivity data relevant to the Glucocorticoid Receptor (GR). Raw data is filtered based on assay type, bioactivity type, confidence score, and organism to ensure data quality and relevance. The curated dataset serves as input for model training.

3.6 Bioactivity of Drugs

To focus the study, the bioactivity of drugs against the Glucocorticoid Receptor (GR) is examined. This receptor plays a significant role in regulating inflammatory responses and is a relevant therapeutic target.

3.7 Assay Type In the initial data collection, Assays are experimental procedures used to measure the
activity of compounds on biological targets.

3.8 Binding Assays (Assay Type B)

Specifically evaluate the binding affinity between a ligand (drug) and its target receptor, in this case, the GR. Binding assays provide quantitative data such as the concentration needed for half-maximal binding (EC50), an important metric for drug potency (de Jong et al., 2005; Tibbitts et al., 2016).

3.9 Bioactivity Type

Standard Name: (Half-maximal Effective Concentration) represents a pharmacological parameter that quantifies the concentration of a compound required to produce 50% of its maximum biological effect. This standardized measurement serves as a critical bioactivity endpoint in drug discovery and development processes.

: EC50 values serve as the primary dependent variable (target variable) for model training and validation. The logarithmic transformation of EC50 values (pEC50 = -log10[EC50]) is commonly employed to:

3.10 Data Quality Considerations:

All EC50 values were standardized to molar (M) or nanomolar (nM) concentrations. Only data from standardized assay conditions were included Preference given to compounds with multiple independent measurements Compound with EC50 values outside detectable ranges were excluded

3.11 Confidence Score

The confidence score evaluates bioactivity data reliability and target specificity (Gaulton et al., 2017).Confidence score ≥ 6 required High scores (≥ 7) ensure single protein interaction**:** Scores reflect experimental methodology reliability For GR receptor studies, high confidence scores ensure compounds primarily interact with GR protein, minimizing off-target effects and enhancing model reliability.

3.12 Organism

3.13 Homo sapiens Selection

All bioactivity data restricted to human (Homo sapiens) sources to ensure clinical relevance and eliminate inter-species variability. Improved data homogeneity for model development

3.13 BAO Format

3.14 Bioassay Ontology Framework

BAO provides standardized vocabulary for biological assay description, enabling systematic data organization and quality assessment. Direct compound-receptor interaction measurement It Downstream signaling pathway activation Cellular response measurements are also important

3.15 Step 2: Data Cleaning

The data cleaning process is an important part in the training of models, to ensure that the accuracy for the machine learning model is not compromised, and the dataset should be precise, and accurate. Here are some steps that are used to modify the data:

3.16 Compound Filtering

The initial dataset typically contains thousands of compounds, including some irrelevant or poor-quality entries. These might be molecules that are chemically unstable, incorrectly annotated, or duplicates. Removing such entries is essential to:

Reduce noise**:** Irrelevant compounds add noise and complexity. Prevent overrepresentation**:** Overabundance of similar compounds can bias the model. Enhance model learning**:** Cleaner data improves pattern recognition and generalization.

This step uses cheminformatics filters and expert knowledge to retain only compounds relevant to the study objectives.

3.17 Ec_{50} Value Filtering

Biological activity values must be consistent and comparable. EC_{50} (half-maximal effective concentration) is a widely accepted measure of a compound's potency. However, bioactivity data may sometimes include other metrics like IC_{50} (inhibitory concentration) or GI_{50} (growth inhibition) or may have missing values.

Reasons for filtering:

- **Standardization:** Keeping only EC_{50} ensures uniformity across the dataset.
- **Reliability:** Missing or inconsistent data reduces model accuracy.
- **Relevance:** EC_{50} directly correlates with biological effect strength.

In practice, this step may lead to the removal of compounds with missing or incompatible bioactivity values, for example, 33 such entries in our dataset.

3.18 Activity Classification Based On Threshold

Machine learning classification tasks often require binary or multiclass labels. For QSAR classification:

- A common cutoff is set at **EC_{50} = 1000 nM**:
    - **Active compounds:** $EC_{50} \leq 1000$ nM
    - **Inactive compounds:** $EC_{50} > 1000$ nM

This threshold reflects pharmacological relevance—compounds below this value are likely to have meaningful biological activity.

3.19 Salt Removal

Many chemical compounds exist as salts or contain counterions, which can interfere with molecular descriptor calculations and modeling. Salts add extraneous atoms and alter physicochemical properties, leading to inconsistent descriptor values.

3.20 Duplicate Removal

Duplicates in the dataset may arise from multiple entries of the same compound in different formats or from data merging. Use chemical structure hashing or fingerprinting

techniques to identify and remove exact duplicates.No duplicates found at this stage indicates a high-quality dataset ready for modeling.

3.21 Pec_{50} Calculation

EC_{50} values span several orders of magnitude and often exhibit a right-skewed distribution, which can challenge regression modeling.

- **Transformation:**

  Using the negative logarithm:

  $pEC_{50}=-\log_{10}(EC_{50})$

3.22 Outlier Removal

3.23 Data Screening and Removal of Missing SMILES

At first the dataset consisted of compounds with missing SMILES (Simplified Molecular Input Line Entry System) notations which are required for computational molecular representation. Since entries without SMILES could not be processed further, they were automatically removed to ensure complete dataset.

This step assures that all compounds have valid molecular representation for descriptor calculation.

3.24 Duplicate Removal by 'Molecule ChEMBL ID'

The **'Molecule ChEMBL ID'** column was scanned to identify exact duplicate entries. Duplicate rows were highlighted and removed to prevent bias or over representation in the model. This process helps maintain data consistency and integrity.

3.25 Filtering Based On Standard Relationship To Ec_{50} Values

Compounds with standard relationship = '=' to their IC_{50} values (as defined by Kalliokoski et al.) were retained. Entries with relations like '>', '<' or approximations were removed to make bioactivity data more precise. By doing so, compounds with exact EC_{50} measurements are the only ones that contribute to model building.

3.26 Removal of Anomalies

Anomalies—outliers or inconsistent data points—were identified and removed. These anomalies could distort the model's predictive power by introducing noise.

The final cleaned dataset consisted of **950 compounds** after anomaly removal. The

Chemical Structures The entire dataset was uploaded to Google Colaboratory as a CSV file and chemical structures were preprocessed using Python scripts. One important step was the removal of salt ions (i.e. fluorine, chlorine, bromine) that appeared in the SMILES notation of compounds. As these salts can negatively impact the calculation of molecular descriptors which are important features in QSAR modeling and reduce the hit discovery efficiency and model prediction accuracy. These salt components standardize the chemical structures so that they are consistent and reliable for subsequent computational analyses (Fourches et al., 2010).

After salt removal and structural standardization, the dataset was screened a second time, this time specifically looking for duplicate compounds found by their SMILES strings. The last step of duplicate removal ensures that there is no duplicate molecule at the structure level in the final dataset so that there is no redundant data that can bias the machine learning models and the integrity of the dataset is maintained for accurate QSAR predictions.

3.27 Conversion Of Ec50 Values To Pec50 Values

A pharmacological measure of great importance, EC50 (half maximal inhibitory concentration) quantifies the potency of a drug to inhibit a specific biological target by half (Rang et al.). IC_{50} values can be reported in units of nanomolar (nM) or micromolar (μM) which can lead to inconsistencies if datasets are integrated. The variability makes data analysis and development of QSAR models more complicated (Kalliokoski et al., 2013). To overcome this, IC_{50} values are converted to Pec50 values by taking the negative logarithm (base 10): pIC50=−log10(IC50)

This transforms the dataset so that it is standardized by compressing the range of values and minimizing skewness. Therefore, the data is made more homogenous which reduces errors and increases the accuracy and robustness of the machine learning models that are trained on the dataset.

3.28 Descriptor Calculation

Molecular descriptors are numerical representations of molecular properties and are the basis for QSAR modeling. Two well established cheminformatics tools were used to calculate descriptor in this study: Molecular Operating Environment (MOE 2022.02) and Mordred. MOE Software: MOE is mainly used for the generation of 3D molecular descriptors because of its advanced energy minimization and conformational sampling capabilities in order to calculate accurate geometric and physicochemical properties. Mordred: This tool works as a complement to MOE computing a large number of 2D descriptors topological, constitutional and physicochemical properties, for a complete description of the compounds. The descriptors were first stored in MOE's proprietary .mdb format for further processing. The descriptor data were converted to the CSV format and imported into Google Collaboratory notebooks for machine learning pipeline integration and preprocessing and feature selection.

3.29 Molecular Descriptor Processing And Dataset Preparation

The molecular descriptors were calculated using two cheminformatics tools: MOE (Molecular Operating Environment) and. These tools are widely recognized for computing a comprehensive range of 2D and 3D descriptors, including physicochemical, topological, constitutional, and geometrical properties (Yap, 2011; Vilar et al., 2012). MOE was specifically used to generate 3D descriptors due to its robust energy minimization and conformational analysis capabilities.

After descriptor calculation, a stringent cleaning process was performed to clean up the data and improve model performance: all descriptors with zero variance (i.e. constant values across all compounds) were removed because they do not contribute to discriminatory power.

Any descriptors with missing values were excluded to prevent introducing bias or needing imputation.

Descriptors were identified as highly correlated with Pearson's correlation coefficient (r) higher than 0.9 and redundant features were eliminated to avoid multicollinearity which can degrade both model interpretability and generalization (Todeschini & Consonni,

2009). The feature selection is conducted in a very meticulous way which in turn increases the robustness, interpretability and predictability of the QSAR models.

3.30 Rule Of Five (Ro5) Violation Assessment

A commonly used guideline for assessing the drug-likeness of compounds, especially their pharmacokinetic properties (absorption, distribution, metabolism and excretion, ADME), is Lipinski's Rule of Five (Ro5) (Lipinski et al., 2001). The rule is that the more criteria that are violated, the less likely compounds are to be orally active drugs.

Molecular weight ≤ 500 Daltons

LogP (octanol-water partition coefficient) ≤ 5

Number of hydrogen bond donors ≤ 5

Number of hydrogen bond acceptors ≤ 10

In this study, compounds that violated more than one of these rules were flagged and excluded to ensure the dataset contained molecules with desirable drug-like properties. This step in filtering allows us to refine QSAR model relevance and practical applicability by considering compounds with good ADME properties.

3.31 Feature Extraction

The relevant molecular descriptors in the data set were chosen as feature for model training after data cleaning and Ro5 filtering. Dimensionality reduction by feature extraction and selection is essential to reduce the number of features, avoid overfitting and improve model performance. While there are techniques such as recursive feature elimination, principal component analysis (PCA) or correlation-based feature selection that can be used to identify the most informative descriptors and remove redundant or irrelevant ones (Guyon & Elisseeff, 2003) such techniques can impose additional costs to the runtime of the learning process. This step improves computational efficiency and model interpretability (particularly important when dealing with high-dimensional descriptor datasets typical for QSAR studies).

3.32 Train-Test Split

The dataset was partitioned into training and testing subsets (typically as 80:20 split) to evaluate the generalizability and robustness of the QSAR model. This stratified splitting makes sure that active and inactive compounds are proportionally present in both subsets and hence will not lead to data imbalance that could bias model evaluation. The model is fit to the training set and its predictive accuracy is assessed on unseen data (i.e. the independent test set) (Hastie, Tibshirani, & Friedman, 2009). This enables to validate the practical utility of the model in drug discovery applications.

3.33 Molecular Docking

In the molecular docking the possible drug candidates that we found with the help of the QSAR model In this process the interaction takes place between hits and proteins and their affinity for each other will tell us about the potency of the drug against the GR receptor

CHAPTER 4

RESULTS

4.1 Data Collection

The initial dataset obtained from BindingDB and ChEMBL databases contained 1,576 compounds, but we extracted the compounds with the bioactivity of EC50, constructing an EC50 subset of 843 compounds. After salt removal, compounds with duplicate SMILES were removed and the dataset was screened throughout to remove outliers. Our final dataset contained 616 compounds, and this dataset was used as our input data.

4.2 Descriptor Calculation And Selection

4.3 MOE Descriptors

A list of 209 2D descriptors was obtained for all 843 compounds using the MOE software.

4.4 Pre-Processing And Descriptor Selection

After descriptor calculation from MOE software, manual preprocessing was performed to remove descriptors with zero or null values. Python was used for the removal of highly correlated features and low-variance compounds and the final **28** descriptors were used for feature selection. These descriptors are as follows.: 'apol', 'arorings', 'a_acc', 'a_don', 'a_hyd', 'a_nO', 'balabanJ', 'h_logD', 'h_logP', 'h_logS', 'h_log_pbo', 'h_pavgQ', 'h_pKa', 'h_pKb', 'h_pstates', 'h_pstrain', 'logP(o/w)', 'logS', 'opr_brigid', 'PEOE_VSA+0', 'PEOE_VSA+1', 'PEOE_VSA+2', 'PEOE_VSA_NEG', 'SlogP_VSA7', 'SlogP_VSA9', 'SMR_VSA1', 'SMR_VSA3', and 'SMR_VSA5'.

The same preprocessing protocol was followed, and these 28 descriptors were used for the unknown dataset as well prior to screening against the QSAR model.

4.5 Feature Selection

4.6 Heat Map

We evaluated the correlation of features, and the following heat map was constructed (see Figure **0.2**), where the values are between -1 and +1. -1 depicts perfect negative correlation and +1 depicts perfect positive correlation.

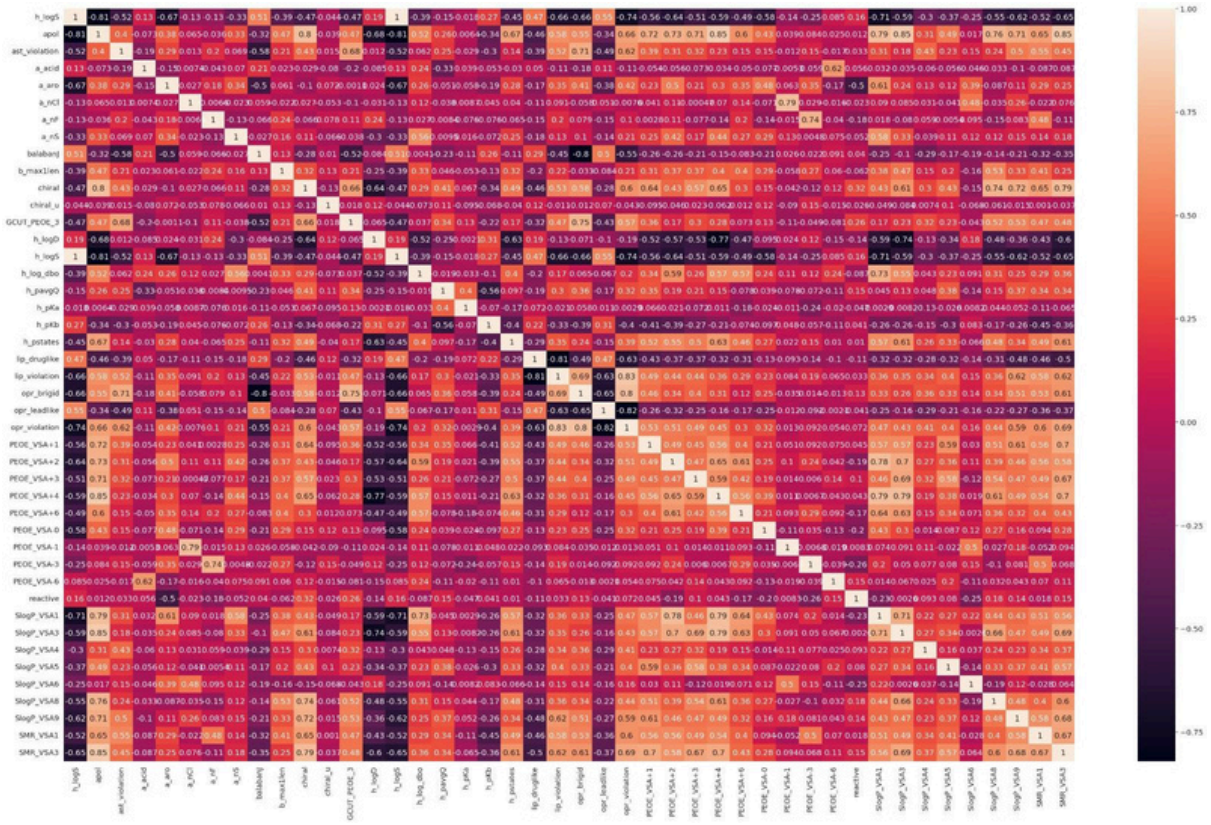


Figure 0.2. Heatmap depictingthe correlation between different 2D descriptors

The Heat Map illustrated a high correlation between LogS, h_logD, and h_logS, in the same way apol and a_don showed the value of 0.78 i.e., they were highly correlated with each other.

4.7 Validation Of Baseline Model

4.8 Rmse Scores

A baseline model was developed using Random Forest Regression, and a mean 10-fold validation was performed. The mean 10-fold validation technique divided our training data into ten equal parts, using nine of these parts to train the model while the tenth was used to validate the model. This process was repeated five times, according to the machine learning algorithm, each time a different part of the data was used as the validation set. The mean of the performance metric across all five validation sets was calculated as Mean 5-fold RMSE score. The random forest regression of the 33% test dataset was also calculated as RMSE test set score. A third RMSE score was obtained by randomizing the target variable in the test set, so it is not the same as those in the training dataset. This evaluated the performance of our model and its robustness using a set of

test data that has had the target variable intentionally randomized or altered. For best fit model, RMSE of training set should be less if it is in zero's than it means error-rate is minimum and that feature explains model properties better and are most important for building model.

A summary of the RMSE scores is provided in the following table.

Table 2. DepictingasummaryoftheRMSEscoresobtainedthroughcross-validation after featureselection

Properties of Validation Models	Obtained Values
Mean 10-fold RMSE	0.30149308
RMSE train set	0.3090022666769

4.9 Filter Method

After the application of the Filter method, constant or quasi-constant features were removed. As a result, 12 features from initial 28 features were left i.e., ' apol', 'a_acc', 'h_logS', 'h_pavgQ', 'h_pKa', 'h_pKb', 'logP(o/w)', 'logS', 'PEOE_VSA+0', 'PEOE_VSA_NEG', 'SMR_VSA1', 'SMR_VSA5'. Mutual information selection helped us identify highly correlated descriptors. Followed by the Mutual Information Selection model, the five most important and five least important features were generated as shown below.

Feature selection of the final 18 descriptors of our dataset gave us a bar graph of the descriptors ranked according to their importance. Descriptor 'LogS' was identified as the most important descriptor for our compounds. This descriptor represents the experimental aqueous solubility of the potential drug compound.

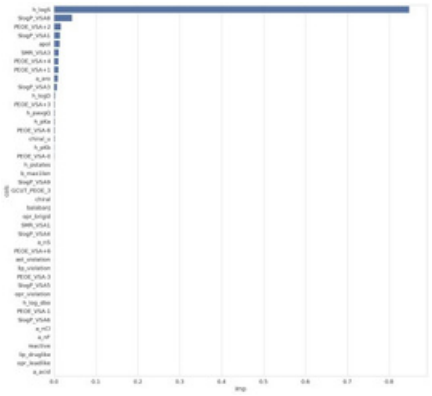


Figure 0.3. Feature selection resulted in a bar plot showing the importance of the 28 descriptors of our dataset

The graph in Figure **0.3** showed the most important feature for building QSAR model was logS followed by h_pavegQ, SMR_VSA5, then SMR_VSA3, and so on.

4.10 Principal Component Analysis

The dataset of 843 compounds was split into 67% train set and 33% test set and further subjected to Principal Component Analysis (PCA). To perform PCA, the dependent Y-variable was defined as pEC50, and an independent X-variable was defined as the principal components.

4.11 Elbow Plot

Elbow plots were obtained for both test and train datasets, these plots (see **Error! R eference source not found.**) show the variance ratio of the set standard 18 PCA components.

The variance ratio of the Test and Train dataset was reported to be 90.04815, respectively. It illustrated that all the PCA components were represented without any loss

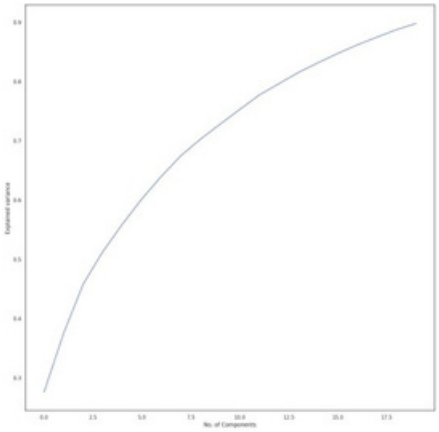


Figure 0.4. Elbow plot of Train and Test Dataset variance explained with 18principal components: A. Variance ratio of Test Dataset, and B. Variance ratio of Test dataset.

4.12 Variability Plots

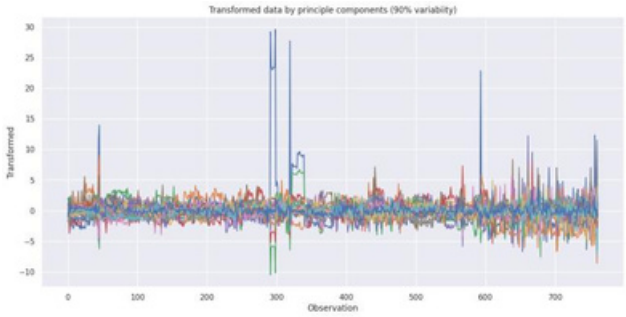


Figure 0.5. Variability Plot of A) Test Dataset transformed by 18 principal components

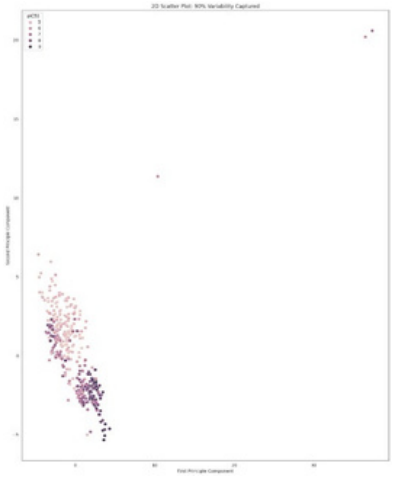
Principal components that preserved 90% variability in test and training datasets were used to plot the variability plot.

The variability plot of Train Dataset gave peak values which illustrated the variation in data defined by 11 principal components in Train Dataset. The variability plot of test dataset also gave us peak values which pointed towards the variation in data that was defined by 12 principal components in Test Dataset (See **Error! Reference source not found.**).

4.13 2d Scatter Plot

Components with 90% variability were used to generate 2D scatter plots of both test and training datasets using the first two principal components of both datasets respectively. The scatter plot of the training dataset provided us with two clusters of values of pIC50 while the scatter plot of the test dataset gave us a single cluster of values of pIC50, highlighting the relationship between the dependent and independent

2D Scatter Plot: 90% Variability Captured for Train Dataset.



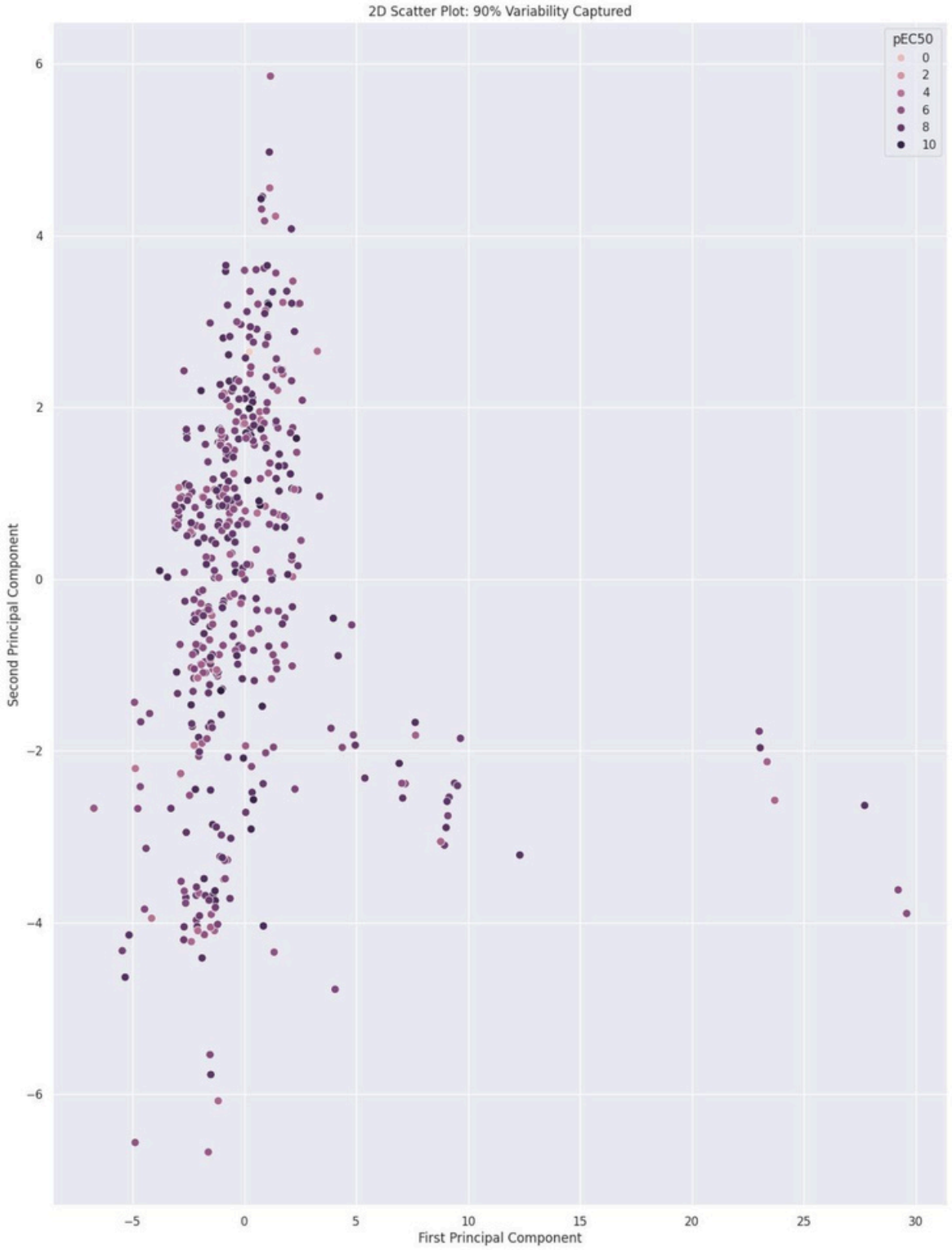


Figure 0.7 2D Scatter Plot: 90% Variability Captured 2D Scatter Pl

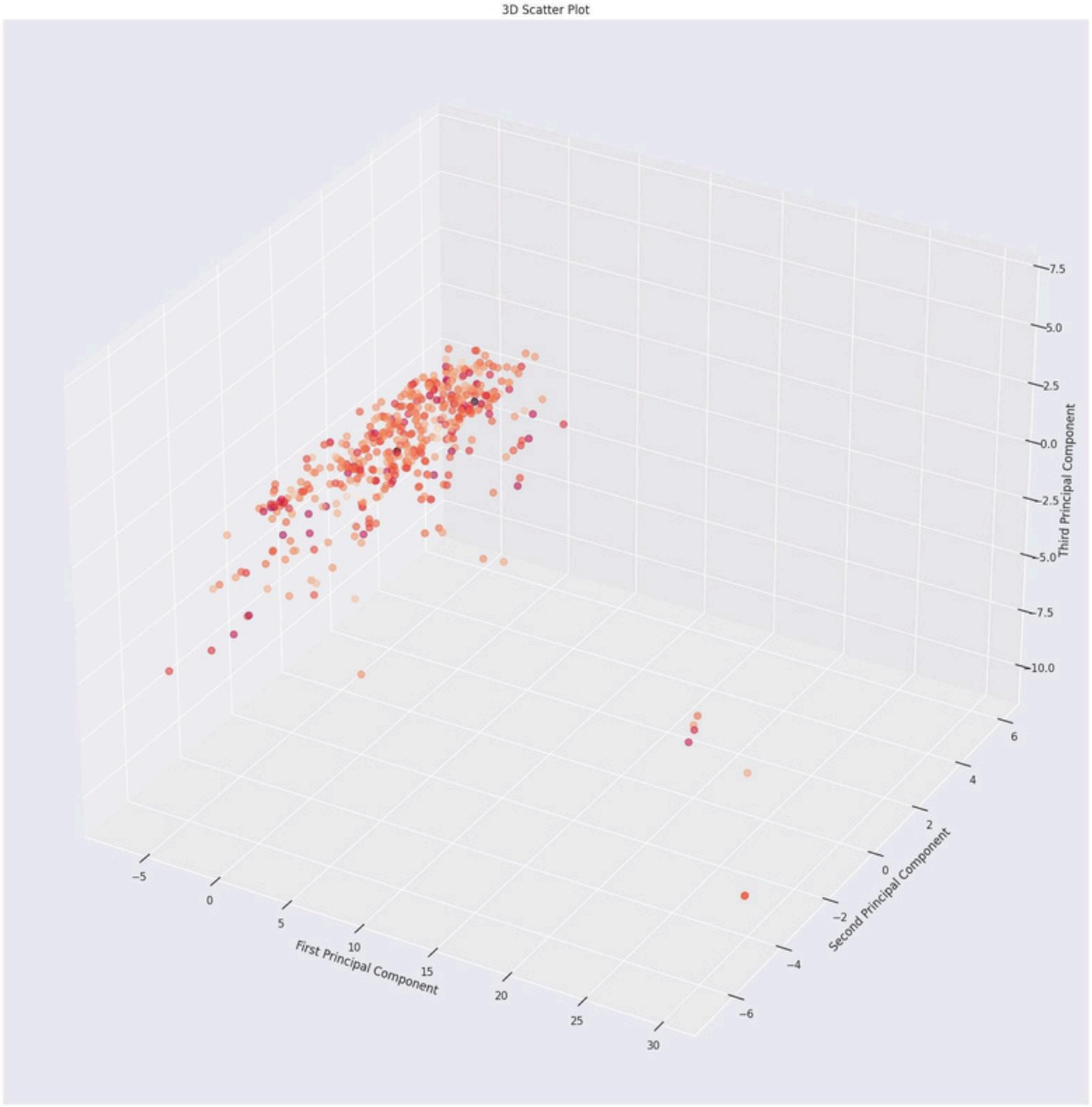
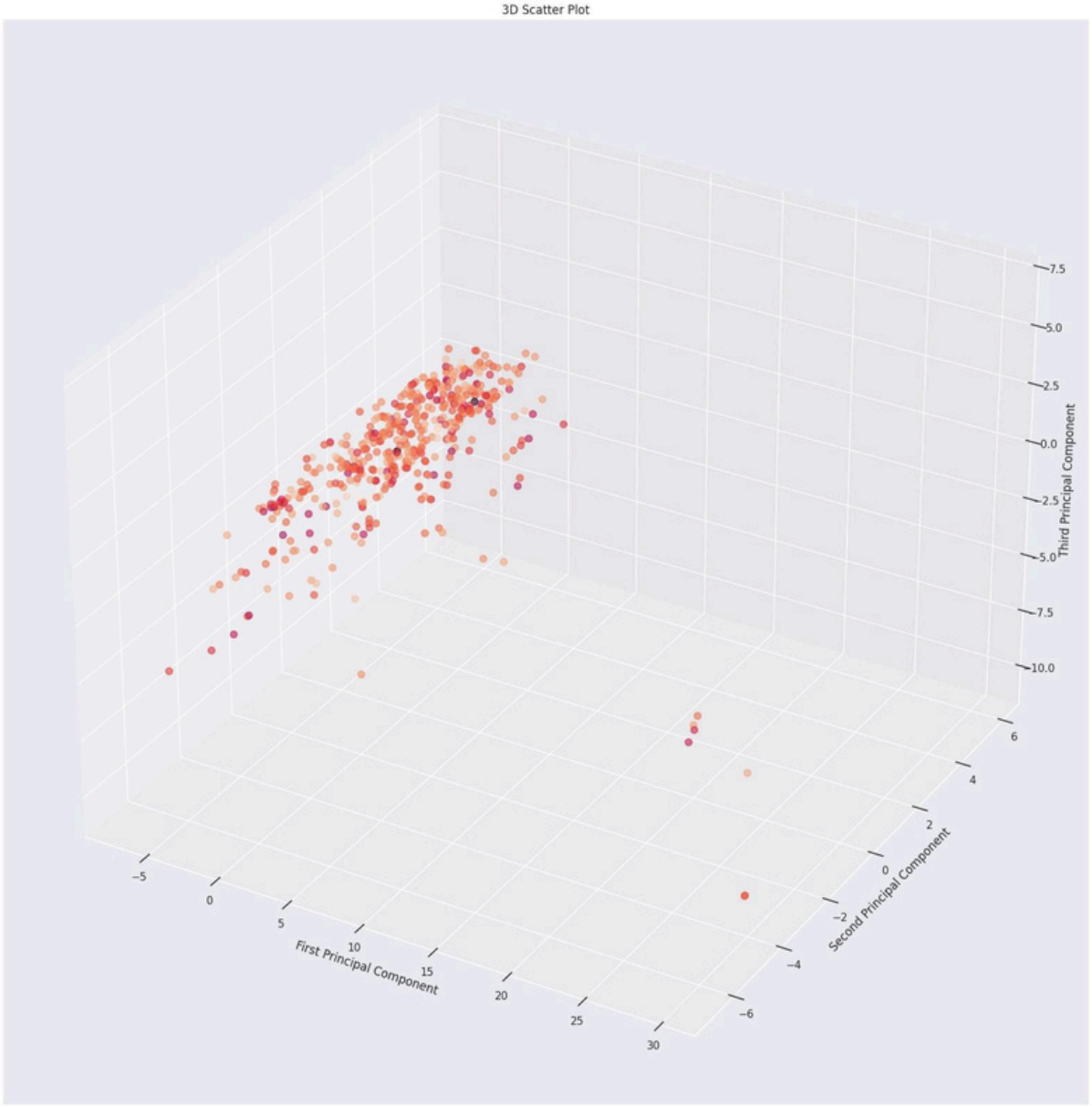


Figure 0.8 3D Plot90% Variability Captured for Test Dataset for Test Dataset 2D ScaPlot:90% Variability Capturedfor Test Dataset4.5.4.

Components with 90% variability were used to generate 3D scatter plots of both test and training datasets using the first two principal components (PC1, PC2, PC3) of both datasets respectively on X-, Y-, and Z- axes. The scatter plot of the training dataset provided us with two clusters of values of pEC50 while the scatter plot of the test dataset gave us a single cluster of values of pIC50, highlighting the relationship between the dependent and independent variables. The 3D scatter plots of the Train and Test datasets showed two clusters of values in the Train dataset and one cluster of values in the Test Dataset which pointed towards the variance of the datasets (see Figures 0.9).



Figures 0.9. (A) and (B) show 3D Scatter plots of PC1, PC2, and PC3 of both Training and Test Datasets Respectively

4.14 Classifiers

For classifiers, we defined X-variables also known as our independent variables as 'PC1', 'PC2', 'PC3', 'PC4', and 'PC5'. Our Y-variables or the dependent variable were defined as the 'Targets'.

Random Forrest Model (MLR), Linear Regression(SVM), and Partial Least Square Regression (PLS) models were used to evaluate our QSAR model and accordingly, values for Train and Test Datasets were

4.15 Experimental Pec50 Vs Predicted Pec50 Linear Plots

We generated a linear plot based on the support vector machine and partial least score model for both training and test datasets respectively. These graphs show that the predicted pEC50 values obtained through RFM models for datasets closely resemble the experimental pEC50 values. Hence, the graphs show the points plotted close to the diagonal line (See **Error! Reference source not found.**). This can be interpreted as a s trong linear correlation between the experimental and the predicted pEC50 values.

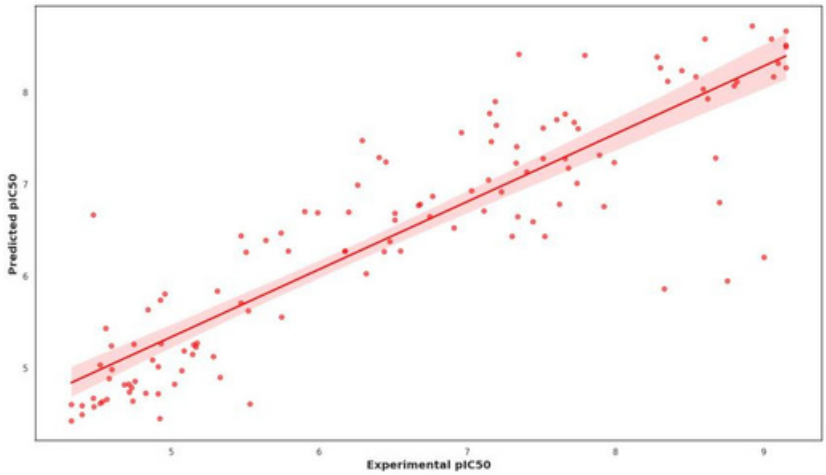


Figure 0.10

4.16 Random Forrest Model

The Random Forrest Model(RFM) Regression Model is a linear regression model that seeks the best-fitting line across a set of data points by minimizing the sum of squared discrepancies between anticipated and actual values. In our case, the data points are the predicted EC 50 and experimental EC50 values of the dependent variable.

RFM regression model generated the following values for Train and Test Dataset. The R-squared value for the train and test data was 0.609 and 0.645 respectively. A model will be considered within the range of moderate to strong fit if the R-squared value is between 0.6 to 0.9. The adjacent R-squared value for the train data was reported as 0.604 and the F-statistic value was reported as 126.6. This F-statistic model shows that the model is statistically significant as a larger F-statistic value is preferred as it indicates the overall significance of the regression model. The adjusted R-squared value ranges from negative to 1 with the values closest to 1 preferred as it indicates that the model is a better fit for the data (Chicco et al., 2021).

4.17 Protein-Drug Interactions

Four hits were obtained after the screening of the unknown test set against our QSAR model. These compounds include 'Vecuronium Bromide', 'Verapamil', 'Cabozantinib', and 'Amiodarone'.

Table3. Themechanismofactionofthecompoundsobtainedasourhits(AdoptedfromPubChemDatabase)

Sr. No.	Compounds
1.	Carbozantinib
2.	Grainestron
3.	Cabozantinib conjugated with Iron oxide Nanoparticles
4.	Grainestron Conjugated with IONP

4.2.1 Protein Structures of Mutant Receptors

The PDB IDs of the selected mutants of estrogen receptor alpha gave us detailed 3D protein structures, which were further used for molecular docking. After ligand removal, we obtained the following 3D protein structures (See **Error! Reference source not f ound.**).

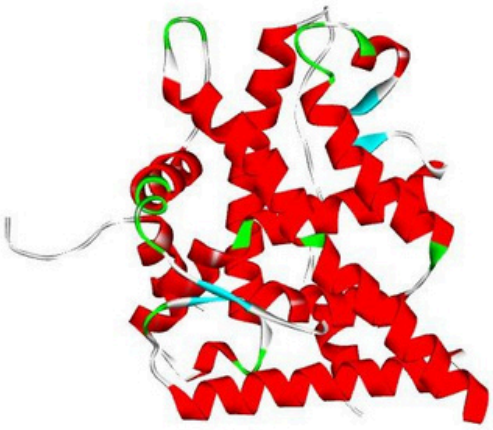


Figure 0.11

4.18 Protein-Drug Interactions

After molecular docking, we obtained the structures of possible binding complexes between our drugs (hits) and protein receptors, and the drug-protein interactions were defined by the resulting docking and confidence scores.

4.19 Docking Models of Carbozantinib

The following binding complexes are obtained when the compound 'Carbozantinib ' is docked on all of our selected proteins. We obtained 2D structures showing bonding between Carbozantinib and our protein receptors. There were two conventional hydrogen bonds observed in Carbozantinib docking models with the wild type and mutant 6SB0 proteins with docking scores -157.01 and -133.23 respectively. The remaining docking models showed no conventional hydrogen bonds but gave significantly high docking scores -157.43 and -154.13 for the compound docking on mutant proteins

CurPocket ID	Vina score	Cavity volume (Å^3)	Center (x, y, z)	Docking size (x, y, z)
C1	-9.6	1442	39, 8, 47	22, 22, 22

Figure 0.12

4.20 Docking Models of Grainestron

The following binding complexes are obtained when the compound 'Grainestron ' is docked on our six of our selected proteins.

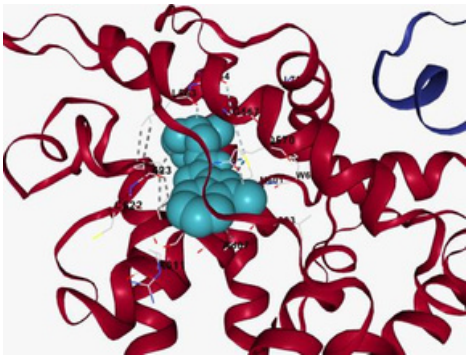


Figure 0.13 The 2D docking models obtained of Grainestron with GR receptor

4.21 Docking And Confidence Scores

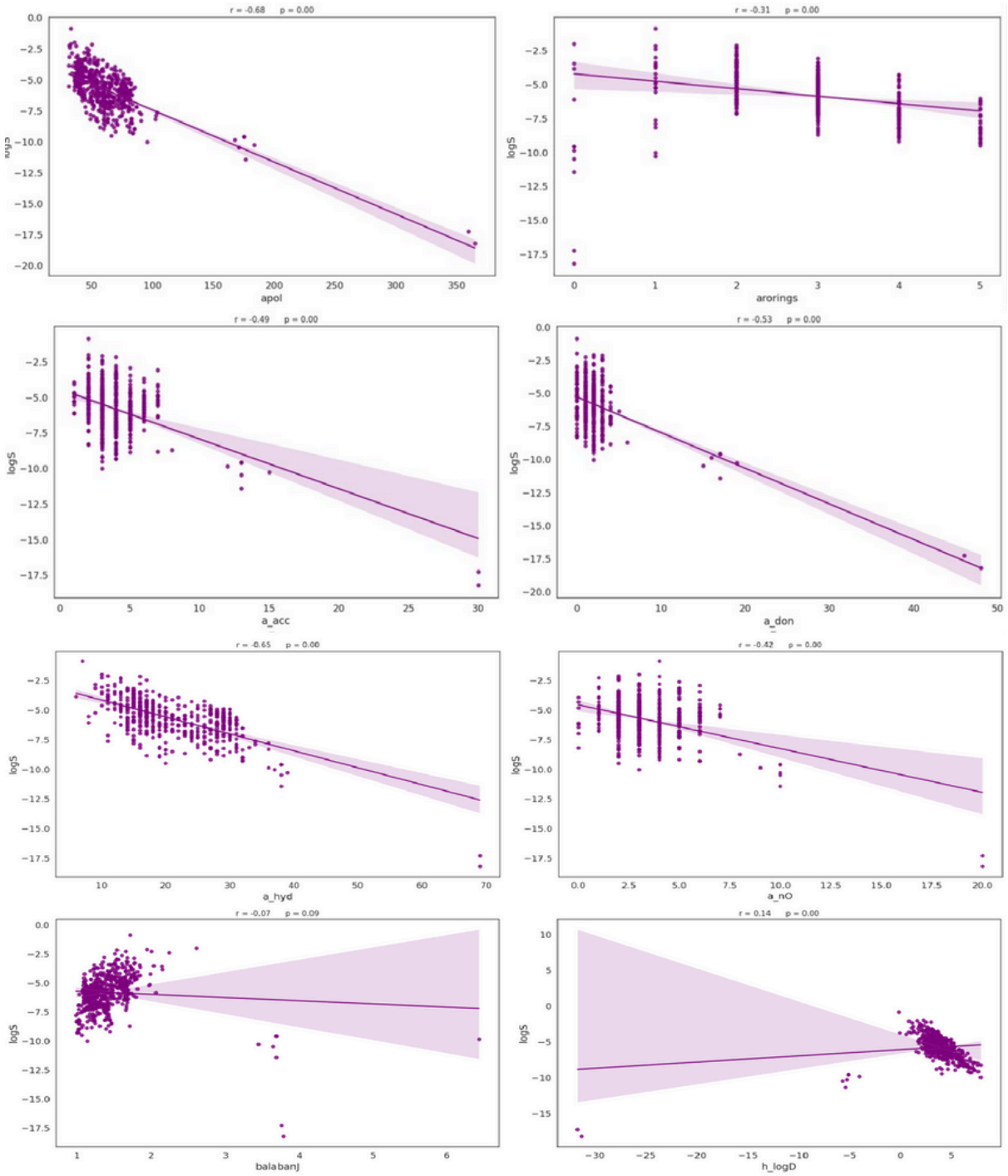
The drug-protein interactions are explained through resultant docking and confidence scores. Typically, when a score is above 0.7, the drug is very likely to bind with the receptor. However, if the obtained confidence score is between the range of 0.5-0.7, there is a possibility that the drug will bind to the receptor. But any confidence score obtained below 0.5 indicates that it is highly unlikely that the drug will bind to the target receptor protein (Yan et al., 2020).

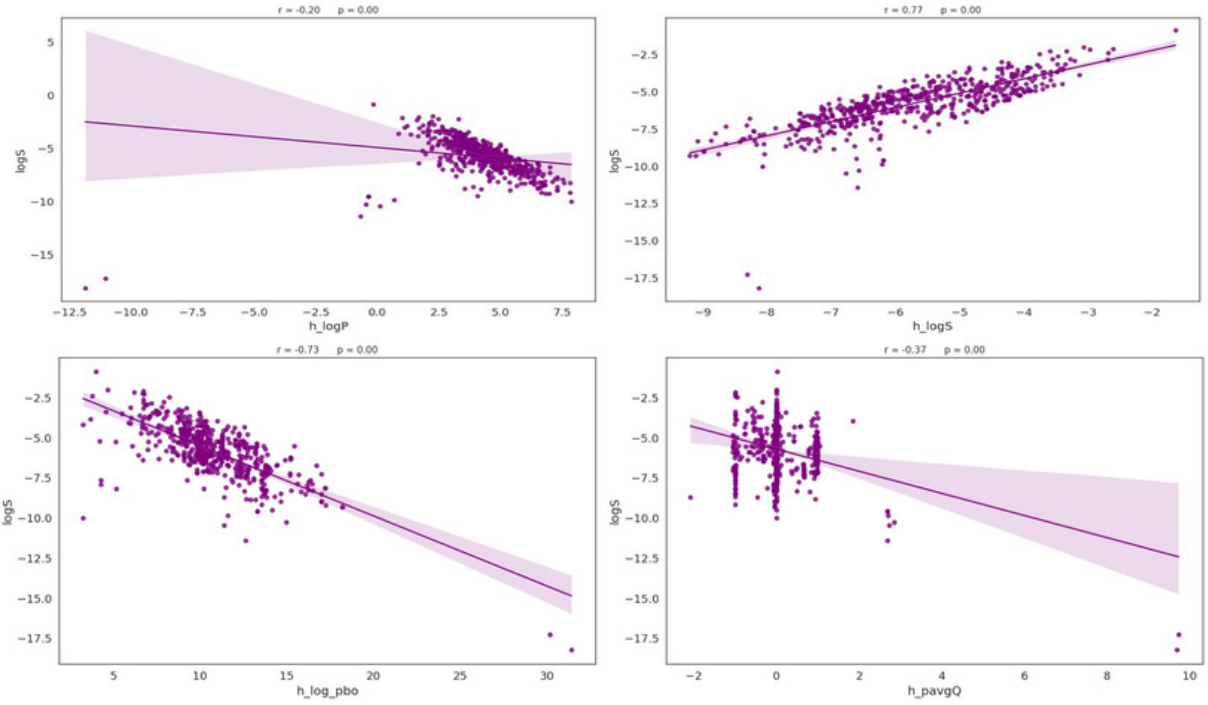
	Name	Visible	Color	Parent	Distance	Category	Types	From	From Chemistry	To	To Chemistry	Angle DHA	Angle HAY	Theta	Theta 2
1	A:UNL1:H...	Yes		Ligand No...	2.41154	Hydrogen Bo...	Conventi...	A:UNL1:H1	H-Donor	A:ME...	H-Acceptor	156.907	90.645		
2	A:GLN57...	Yes		Ligand No...	3.61403	Hydrogen Bo...	Pi-Donor ...	A:GLN57...	H-Donor	A:UNL1	Pi-Orbitals			8.796	
3	A:GLN57...	Yes		Ligand No...	3.90708	Hydrogen Bo...	Pi-Donor ...	A:GLN57...	H-Donor	A:UNL1	Pi-Orbitals			23.916	
4	A:MET60...	Yes		Ligand No...	4.85346	Other	Pi-Sulfur	A:MET60...	Sulfur	A:UNL1	Pi-Orbitals			82.193	
5	A:ALA60...	Yes		Ligand No...	4.7543	Hydrophobic	Amide-Pi ...	A:ALA60...	Amide	A:UNL1	Pi-Orbitals			29.658	36.218
6	A:ALA60...	Yes		Ligand No...	3.90238	Hydrophobic	Amide-Pi ...	A:ALA60...	Amide	A:UNL1	Pi-Orbitals			18.945	10.664
7	A:MET60...	Yes		Ligand No...	4.46546	Hydrophobic	Alkyl	A:MET604	Alkyl	A:UNL1	Alkyl				
8	A:ALA60...	Yes		Ligand No...	3.93418	Hydrophobic	Alkyl	A:ALA607	Alkyl	A:UN...	Alkyl				
9	A:UNL1 - ...	Yes		Ligand No...	4.58133	Hydrophobic	Alkyl	A:UNL1	Alkyl	A:LE...	Alkyl				
10	A:UNL1:C...	Yes		Ligand No...	5.11435	Hydrophobic	Alkyl	A:UNL1:...	Alkyl	A:ME...	Alkyl				

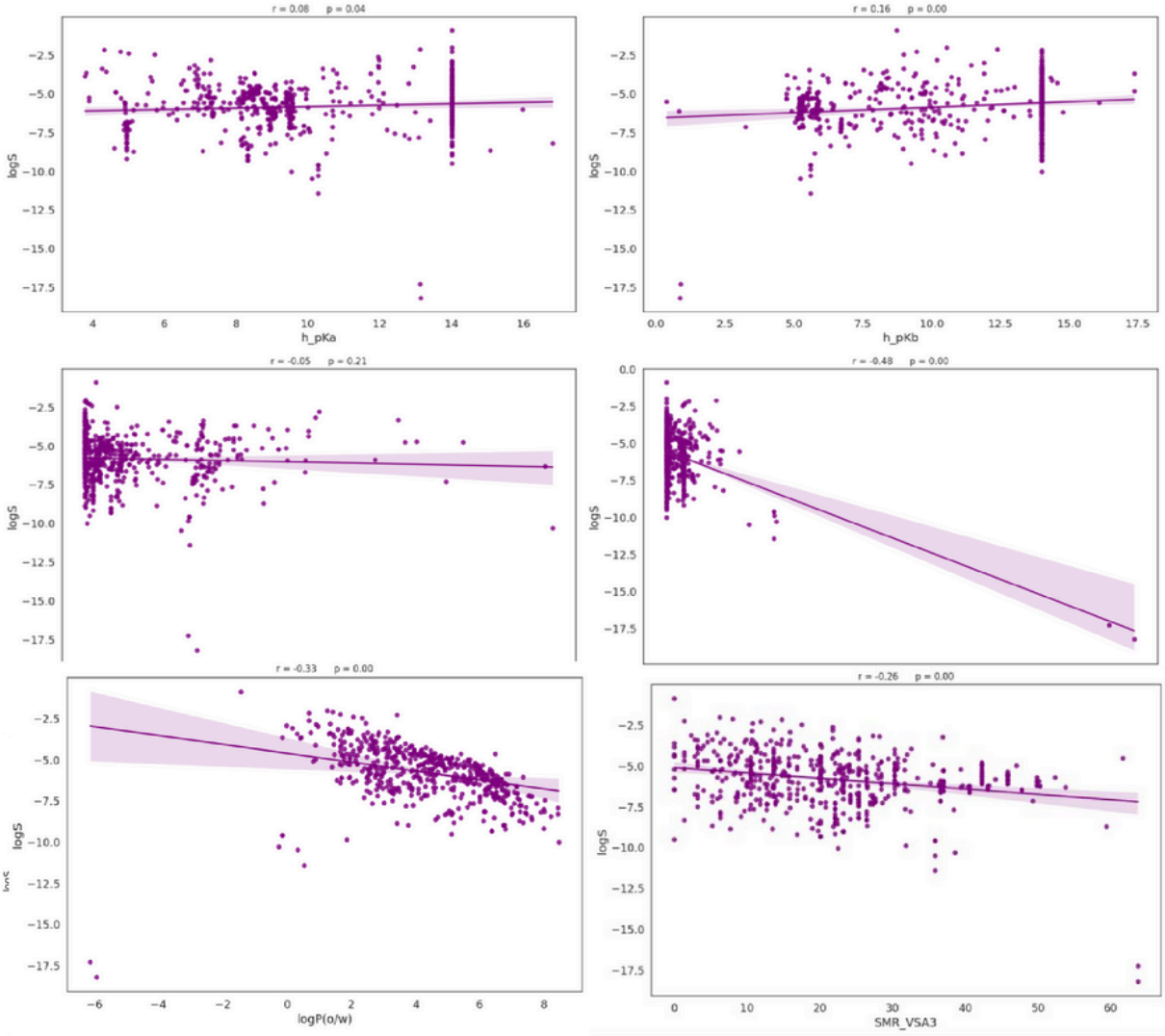
Figure 0.14

4.22 Regression Models

The following is a list of regression scatter plots of our descriptors against 'logS' or the descriptor presenting solubility obtained during feature selection.




r = -0.20 p = 0.00
logS
h_logP
r = 0.77 p = 0.00
logS
h_logS
r = -0.73 p = 0.00
logS
h_log_pbo
r = -0.37 p = 0.00
logS
h_pavgQ


r = 0.08 p = 0.04
logS
h_pKa
r = 0.16 p = 0.00
logS
h_pKb
r = -0.05 p = 0.21
logS
r = -0.48 p = 0.00
logS
r = -0.33 p = 0.00
logS
logP(o/w)
r = -0.26 p = 0.00
logS
SMR_VSA3

DISCUSSIONS

Breast cancer is classified as the most common cause of deaths among malignant tumor diseases. This project aimed to explore drug repurposing *via* machine learning models such as QSAR modeling. QSAR model was built on pEC50 values through the series of steps mentioned in the methodology. We chose to build on pEC50 values, because unlike EC50 values, these values do not have units therefore models built on such values can be interpreted and validated with reduced possibility of error. One of the intentions of our project was to find potential drug compounds that can be tested based on their aqueous solubility in *in vitro* experiments, therefore we obtained a dataset which could translate into molecular descriptors related to solubility (logS). Experimentally determined aqueous solubility of a compound is an important factor in determining the efficacy of the drug compounds (Sorkun et al., 2019).

To ensure our dataset could be later used for wet lab testing, feature selection was performed with logS being set as the standard. Seaborn RegPlot, Heat Map, Filter Method, Wrapper based Method, Intrinsic Methods and RMSE scores (Goodarzi et al., 2012; Kausar & Falcao, 2018) were used to validate the selected dataset for logS.
Seaborn RegPlot was used to represent the correlation between the selected descriptor (LogS) and the target descriptors. The line through the graphs represented the linear regression. The direction of the slop showed negative or positive correlation of the two variables. The clustering of the data around the regression line depicted the strength of the relationship and the scattering of the data points around the line was used to find the variability.
Results of Seaborn RegPlot were summarized in a bar graph, listing out the features based on their importance for developing a good QSAR model. The most important feature was logS with a value greater than that of 0.8 on the importance scale.

Heat Map is also used to represent the correlation between two variables, its values range from -1 to +1 (Sheridan, 2019). The Heat Map highlighted high correlation between different features such as that of high positive correlation between logS and h_logS i.e., 0.77 depicts that increase in hydrophobicity will increase with aqueous solubility, in the same way the positive correlation between logS and h_logD illustrates that increase in lipophilicity will increase as the aqueous solubility increase or *vice versa*. Other features

which showed high correlation with each other includes apol and a_don with the value of 0.78. These features ultimately indicate how much information about lipophilicity, solubility, polarizability etc. is present in our dataset and their positive correlation show how much the compound's chemical properties such as solubility affects its pharmacokinetics and pharmacodynamics.

To further cross validate our datasets, a Mean 5-fold RMSE score of a baseline model was calculated to be 0.2023. This also recorded RMSE value of the Test and Train dataset. Low scores of Mean 5-fold RMSE and RMSE test set indicates insignificant difference between the model's predicted values and actual values of the data (Kausar & Falcao, 2018). Filter Method also identified logS as the most important feature with the value of 4.718 and the list of the features obtained *via* Wrapper Based Method also included logS as one of the selected features to increase the performance of QSAR model. Feature selection is a step to reduce computation time and storage, remove noisy data to prevent overfitting (Kausar & Falcao, 2018). Lastly to validate the final selected features after the application of Random Forest Model and LASSO regularization, an RMSE test score of 0.30900 which highlights very small predictive error (Kausar & Falcao, 2018). After the cross validation of the datasets, using the PCA (Goodarzi et al., 2012), a preliminary test, a variance ratio of Train and Test dataset with 11 and 12 PCA components respectively was generated to be 98% and 97.946% to maximize the separation between the datapoints and reduce the dimensionality. Furthermore, 2D and 3D scatter plot of Train and Test have clusters of data points. Overlapping clusters highlight the similarity in datapoints between the components. At the same time the distances between molecules reflected similarity between the values of the selected PCA components (Ebeling et al., 2013). In the train set scatter plot region of overlapping clusters is small but it has distinct points (Roy Choudhury et al., 2013). The observed similarities and differences between the distances between compounds in the respective scatter plots indicate the relationships and the underlying patterns within the data which can be used improve our model's performance. The overlapping clusters and distinct points in the scatter plots have been analyzed to optimize the accuracy, reliability and predictive power of the model.

The machine learning built based on PLS was classified *via* validation model of MLR and SVM, where PLS and SVM generated same values for the Train and Test sets, i.e., 0.804 and 0.814 respectively (Thakur et al., 2022; van de Waterbeemd, 2008). Providing sufficient validation for the created model. The linear plot, generated from both algorithms to check the predictive ability of the QSAR model, showed very similar values for predicted pEC50 and experimentally determined pEC50 values further highlighting the accuracy of the predictability of the model.

A statistical method, OLS, was used to further analyze the cross validity of the model. The higher R2 and F values with lower RMSE values highlights the predictive ability of the model (Khan & Roy, 2018; Pourbasheer et al., 2019). According to a study (Ozili, 2023), R2 value falling between 0.50-0.90 is considered significant, resulting in best fit model. The generated values of R2 (0.609), Adj R2(0.604), and F-statistic (126.6) for the train datasets and R2 (0.645), and Adj R2(0.636) are considered optimal for both sets, indicating an appropriate predictive ability of the built model (Thakur et al., 2022). To meet one of our aims, i.e., repurposing of the drug compounds for treatment, the model was tested on unknown FDA approved drug compounds to find potential drugs (hits) for agonist against GR. This was followed by molecular docking of mutant receptors against the hits obtained to validate our model.

The docking score, confidence score and conventional H-bonds are considered the three most important features in the docking results. More negative docking score along with high binding energy/high confidence score is preferred as it increase the possibility of binding between the drug hits and the mutant receptors (Ilaghi-Hoseini & Garkani-Nejad, 2022b). Conventional H-bonds are the most stable bonds so a drug-receptor complex having one or two Hydrogen bonds is preferable. Our results further validate our model performance as all the hits that were obtained have high possibility of binding to the chosen mutant receptors but if we were to select the most stable drug-receptor complex that would be Cabozantinib, because it formed complex with nearly all mutated protein receptors with relatively higher negative docking score, higher confidence score, and at most all the formed complex has minimum one conventional H-bond in them, except for Cabozantinib-5W98 model. Followed by Cabozantinib, Vecuronium Bromide also gave optimum scores for docking score, confidence score and conventional H-bond. In

comparison Amiodarone and Verapamil drugs are predicted to be relatively poor inhibitors of ER protein receptors.

By constructing a QSAR model based on their experimental data, our team was able to effectively predict the bioactivity of unknown compounds. Our QSAR model made these predictions using a collection of chemical descriptors, and its validation findings demonstrated that it was highly predictive. We were able to assess the model's effectiveness by comparing its predictions to the actual bioactivity values of a test set of compounds that had not been employed in the model's development. This validation approach demonstrated that the model could accurately predict the bioactivity of these compounds. We also utilized our QSAR model to find pharmacological hits with potential binding and agonist activity for GR. These hits were chosen based on their projected bioactivity, and we next investigated their binding potential to the target protein using molecular docking experiments. These investigations' high docking scores supported our model's prediction capabilities and offered more evidence of its ability to forecast the bioactivity of unknown compounds.

CONCLUSION AND FUTURE PROSPECTS

In conclusion, our project aimed to explore drug repurposing for potential agonist candidates against GR through machine learning model building. We built the QSAR model on pEC50 using the steps explained in the methodology. This machine learning model was built on partial least square (PLS) regression and later classified *via* validation models such as support vector machine (SVM) and multiple linear regression (MLR). Overall, our QSAR model has shown considerable promise in accurately predicting chemical bioactivity and finding prospective therapeutic hits. Docking of the obtained hits on the five mutant and one wild type ERα proteins gave promising results. Our model's high prediction power and confirmation data from molecular docking studies make it an innovative tool for drug discovery and development. Therefore, we accomplished all three of our main objectives.

We aim to optimize the performance of our QSAR model by incorporating more descriptors and features relevant to estrogen receptor alpha. Furthermore, the machine learning process can be utilized to incorporate proteomics and genomics data that can give us comprehensive understanding of novel pathways involved in drug response, in the light of endocrine resistance in carcinogenesis. Pose analysis and molecular docking studies can be expanded to understanding the binding interactions of our hits against the mutated estrogen receptors. We also aim to test these compounds *via in vitro* testing which can further validate their potential use in targeted breast cancer treatments.

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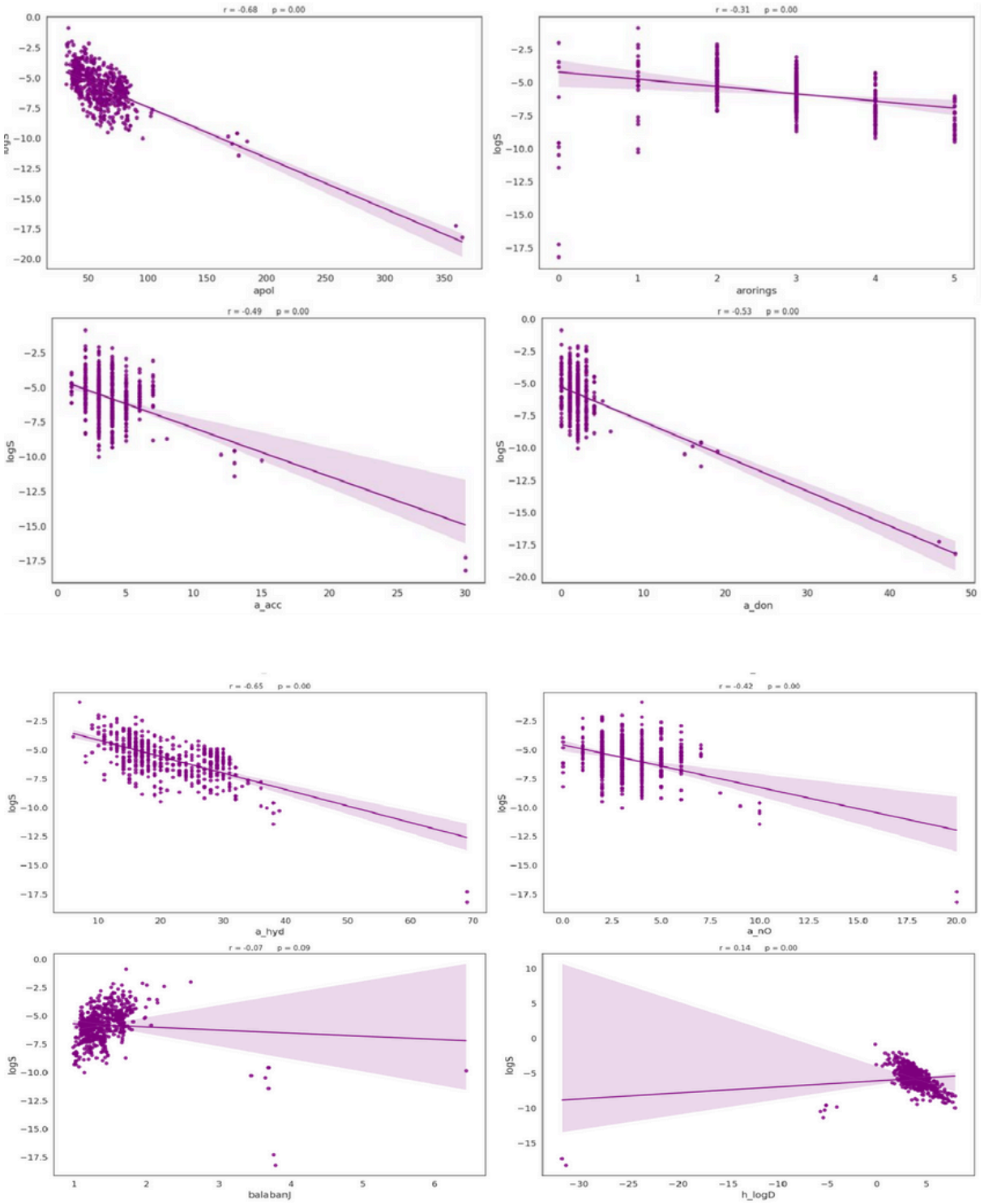
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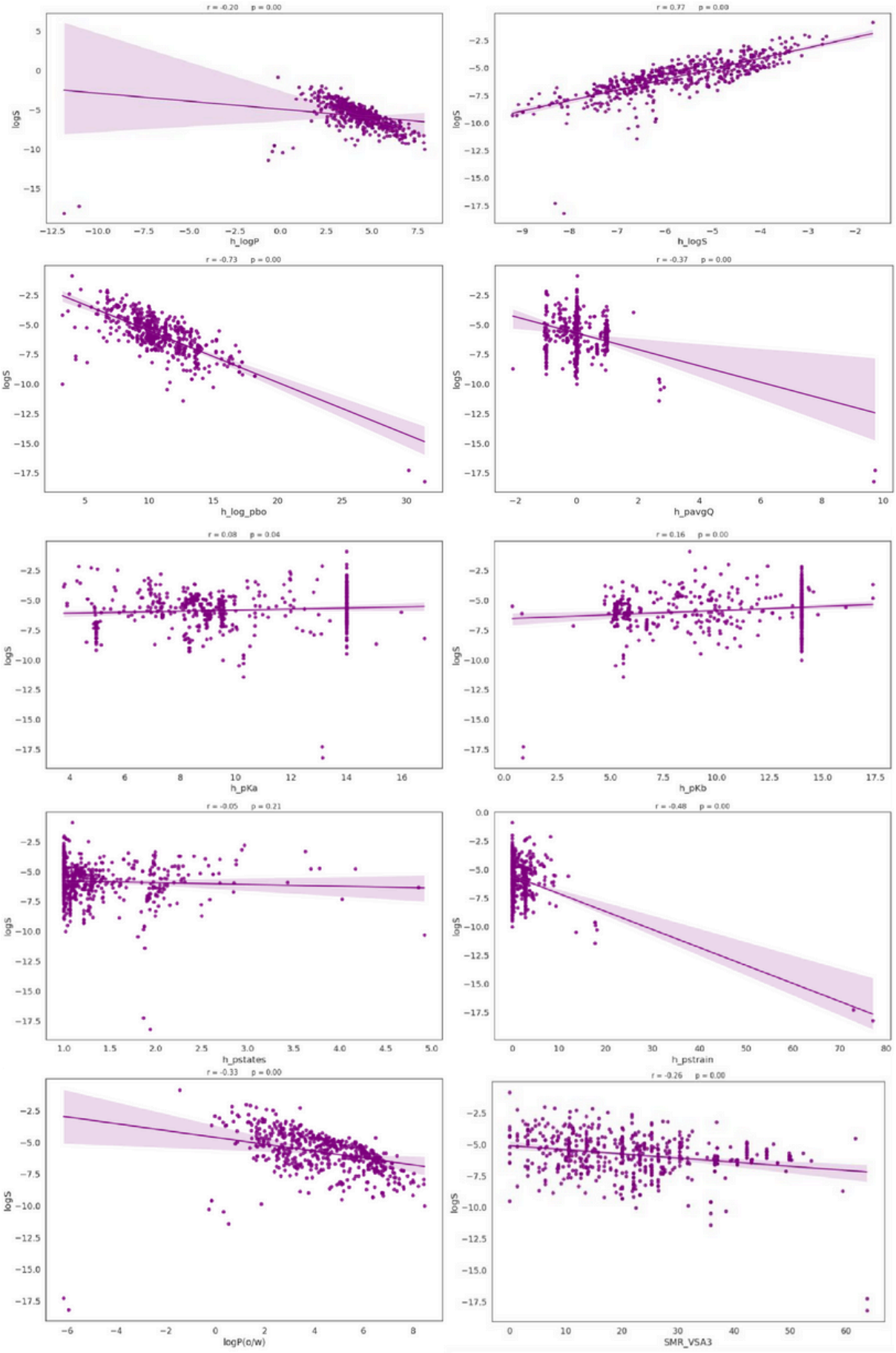
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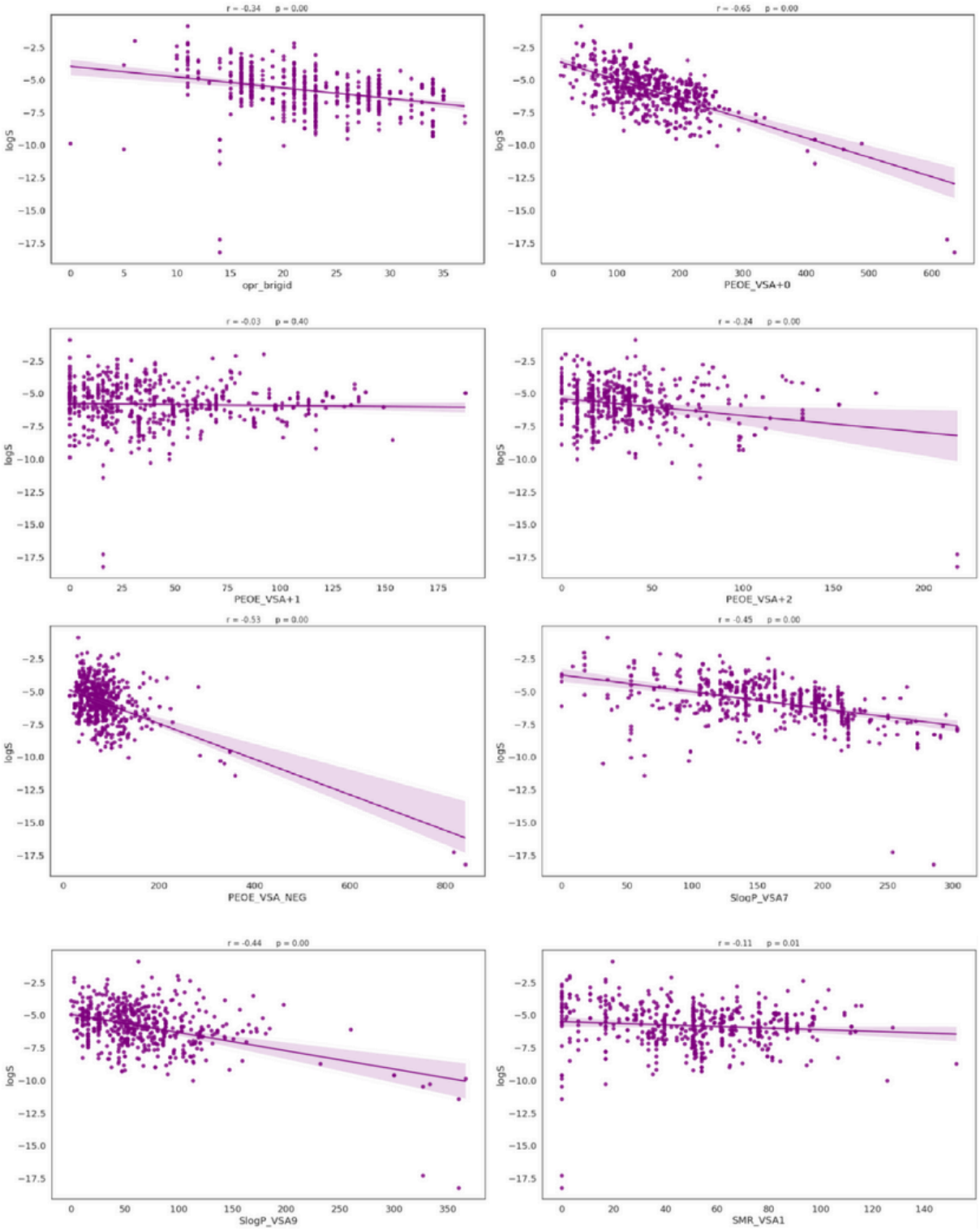
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APPENDIX A

The following is a list of regression scatter plots of our descriptors against 'logS' or the descriptor presenting solubility obtained during feature selection.




r = -0.20 p = 0.00
logS
h_logP
r = 0.77 p = 0.00
logS
h_logS
r = -0.73 p = 0.00
logS
h_log_pbo
r = -0.37 p = 0.00
logS
h_pavgQ
r = 0.08 p = 0.04
logS
h_pKa
r = 0.16 p = 0.00
logS
h_pKb
r = -0.05 p = 0.21
logS
h_pstates
r = -0.48 p = 0.00
logS
h_pstrain
r = -0.33 p = 0.00
logS
logP(o/w)
r = -0.26 p = 0.00
logS
SMR_VSA3


r = -0.34 p = 0.00
logS
opr_brigid
r = -0.65 p = 0.00
PEOE_VSA+0
r = -0.03 p = 0.40
PEOE_VSA+1
r = -0.24 p = 0.00
PEOE_VSA+2
r = -0.53 p = 0.00
PEOE_VSA_NEG
r = -0.45 p = 0.00
SlogP_VSA7
r = -0.44 p = 0.00
SlogP_VSA9
r = -0.11 p = 0.01
SMR_VSA1

APPENDIX B

Our project completion yielded the following Google Colab notebook as a result: https://colab.research.google.com/drive/1mPCg5zTxHFk3zMrL_pj_9KM3SoqDi-_9?usp=sharing.